MBA FINAL YEAR PROJECT REPORT

Innovation of Financial Services:

Navigating the Future of Banking and Investment

A Case Study on Razorpay

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Submitted by:

Supriya Jadhav

Under the Guidance of:

Dr. Vandana Sivaraj

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Abstract

The financial services sector is undergoing a transformation of a kind and speed that has no historical precedent. The convergence of mobile-first consumer behavior, open-source technology infrastructure, cloud economics, and regulatory liberalization has dismantled barriers that once protected incumbent banks from competitive challenge. In their place has emerged a new class of technology-native financial services provider   firms that deliver banking, lending, payments, and insurance through application programming interfaces (APIs), machine-learning models, platform ecosystems, and now autonomous AI agents embedded into everyday digital experiences.

This dissertation investigates that transformation through two lenses. The first is wide-angle and macro: it examines the theoretical foundations of financial innovation, the structural dynamics of India's digital payments ecosystem (UPI processing 22.35 billion monthly transactions by April 2026, representing 50% of global real-time digital transaction volume), the regulatory architecture governing fintech activity (updated through May 2026 to incorporate the Digital Lending Directions 2025 and DPDP Act 2025), and the global practitioner literature on embedded finance and platform economics. The second lens is narrow and case-focused: it trains a detailed analytical spotlight on Razorpay   India's leading B2B payment infrastructure company, now serving over 12 million businesses and processing approximately USD 180 billion in annual Total Payment Volume   as the most instructive global example of how a technology company can progressively absorb, automate, and redistribute functions once exclusive to licensed financial institutions.

The research adopts a mixed-methods design. Qualitative analysis draws on structured literature synthesis, regulatory document analysis, framework application (PESTLE, Porter's Five Forces, SWOT, Disruptive Innovation Theory, Platform Economics), and systematic analysis of Razorpay's product suite through May 2026. Quantitative analysis synthesises market-size data, disclosed performance metrics (Rs. 3,783 crore FY25 revenue; 65% YoY growth), and industry benchmarks from IMARC Group, RBI, PwC, McKinsey, and NPCI.

Key findings establish four mutually reinforcing pillars of Razorpay's competitive architecture: an API-first infrastructure that creates deep developer adoption and measurable switching costs; a transaction-data advantage that enables embedded finance products to serve SME credit markets traditional institutions structurally cannot reach; an AI-powered automation suite that deepens operational integration culminating in the March 2026 Agent Studio launch (autonomous AI agents for disputes, payment recovery, and cash-flow forecasting) and the April 2026 OpenAI Codex MCP Server integration (payment infrastructure setup in under five minutes); and a proactive regulatory compliance posture converting licensing obligations into institutional barriers. Provider-level findings are interpreted cautiously because growth is churn-sensitive.

The dissertation introduces and updates the DELTA Model of FinTech Competitive Advantage (Data-Embedded Finance-Lock-in-Technology-Advancement), extending it with a Stage A+ (Agentic Intelligence) layer that reflects the qualitative shift from passive infrastructure to active financial intelligence observed in Razorpay's 2025–2026 evolution. Strategic recommendations are addressed to four stakeholder groups: incumbent financial institutions, FinTech firms, SME business users, and regulatory architects updated to reflect the confidential IPO filing (April 2026, USD 600–700M, USD 5–6B valuation), international expansion into 25+ countries, and India's trajectory toward 379 billion annual UPI transactions by FY2026–27.

Keywords: *Financial Innovation, Razorpay, Embedded Finance, API Banking, Agentic AI, UPI, DELTA Model, FinTech India, Disruptive Innovation, Platform Economics, SME Finance, Agent Studio, MCP Server, OpenAI Codex, RazorpayX, Open Banking.*

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List of Abbreviations

Abbreviation	Full Form
AA	Account Aggregator
AI	Artificial Intelligence
API	Application Programming Interface
B2B	Business-to-Business
BaaS	Banking as a Service
BCG	Boston Consulting Group
BNPL	Buy Now Pay Later
CAGR	Compound Annual Growth Rate
CBDC	Central Bank Digital Currency
CEO	Chief Executive Officer
CFO	Chief Financial Officer
DELTA	Data–Embedded Finance–Lock-in–Technology–Advancement
DPDP	Digital Personal Data Protection Act
EMI	Equated Monthly Instalment
FDI	Foreign Direct Investment
FY	Financial Year
GDP	Gross Domestic Product
GMV	Gross Merchandise Value
IPO	Initial Public Offering
KPI	Key Performance Indicator
KYC	Know Your Customer
LLM	Large Language Model
MCP	Model Context Protocol

Abbreviation	Full Form
MDR	Merchant Discount Rate
ML	Machine Learning
MSME	Micro, Small and Medium Enterprise
NBFC	Non-Banking Financial Company
NPA	Non-Performing Asset
NPCI	National Payments Corporation of India
OCEN	Open Credit Enablement Network
PA	Payment Aggregator
PCI DSS	Payment Card Industry Data Security Standard
PESTLE	Political, Economic, Social, Technological, Legal, Environmental
RBI	Reserve Bank of India
SDK	Software Development Kit
SEBI	Securities and Exchange Board of India
SME	Small and Medium Enterprise
SWOT	Strengths, Weaknesses, Opportunities, Threats
TPV	Total Payment Volume
UPI	Unified Payments Interface
USD	United States Dollar
VAN	Virtual Account Number

CHAPTER 1

Introduction

1.1 The Transformation of Financial Services   A Global Perspective

The evolution of financial services can thus be said to be an evolution of information processing capabilities. Banks have always been places where information gathering, assessing, and actions based on information concerning borrowers, depositors, and counter-parties have been taking place. It is the technology that enables all that information gathering, assessment, and acting that has been developing at such an accelerated pace from the early 2010s onward and not the process itself.

The years from 2009 to 2026 saw the rise of FinTech 3.0 – an evolutionary path that did not involve improvements to the existing banking process but the disaggregation of the financial services chain into separate pieces each one offered by a different entity via APIs. This was a radical departure from the traditional model of a bank which combined the services of accepting deposits, offering loans, providing payment services, handling foreign currency transactions, and offering investment advice within its single framework. Now, with 2026 around the corner, FinTech 4.0, characterized by agentic artificial intelligence-based systems, is becoming apparent.

Such changes are quantifiable in nature. The global fintech market size is estimated to be USD 460.76 billion in 2026, achieving 18.2% CAGR to reach USD 1.76 trillion in 2034 (Fortune Business Insights, 2026). Global investments in the fintech sector that had surged to USD 210 billion in 2021 but slowed down to about USD 113 billion in 2023 have made their comebacks. In 2025 alone, India has raised USD 3.4 billion from fintech VC investment, taking the third position on the global stage behind the USA and the United Kingdom (Innovate Finance, 2025). Digital payments surpassed USD 3.4 trillion in transaction values within India's UPI payment mechanism alone in 2025 (NPCI, 2026); embedded financial services products have been incorporated into platforms with more than 1.2 billion customers globally (McKinsey, 2023); and alternative lenders using machine-learning underwriting extended hundreds of billions in SME credit annually, credit that traditional banks would not have originated.

As of May 2026, a new frontier has emerged: agentic AI systems capable of autonomously managing dispute responses, recovering failed payments, forecasting cash flows, and through Razorpay's integration of UPI into OpenAI’s Codex platform allowing any developer to embed the full spectrum of Indian payment methods into a new application in under five minutes. These developments represent not merely incremental fintech innovation but a qualitative shift in the human-machine division of financial labour.

1.2 India as a Laboratory of Financial Innovation

Within the international context of fintech evolution, India enjoys a uniquely analytical position. There is no other nation, given similar developmental challenges, with an estimated 1.4 billion citizens, millions of rural families, 22 recognized languages, and per capita GDP a fraction of OECD nations, that has built such a robust digital finance system.

Three policy interventions stand out as transformative. The first was the Aadhaar biometric identity system, which by 2025 had enrolled 99.7% of India's adult population and provided a universal, API-accessible identity layer that removed the foundational barrier to digital financial inclusion, the inability to prove identity for a population historically excluded from formal financial systems precisely because of this barrier.

The second intervention was the November 2016 demonetization exercise, which whatever its macroeconomic controversies served as a forced inflection point for digital payments adoption. Overnight, 86% of India's currency in circulation was rendered legally invalid, creating emergency demand for digital payment alternatives that proved behaviorally durable.

The third and analytically most important step taken by NPCI was the introduction of Unified Payments Interface (UPI) from August 2016. UPI is a real-time, interoperable and zero-MDR payment interface that allows real-time transfers between banks using virtual payment address on any mobile phone. The third and analytically most important step taken by NPCI was the introduction of Unified Payments Interface (UPI) from August 2016. UPI is a real-time, interoperable and zero-MDR payment interface via virtual payment addresses using any smartphone. Its design decisions interoperability between all participant banks, zero consumer charges, and open API access for licensed payment service providers reflected a deliberate policy choice to build payment infrastructure as a public good.

They speak for themselves, and the trends are only getting more exciting. By 2025, UPI had handled over 250 billion transactions amounting to about $3.4 trillion, forming 50% of total global volume of transactions in real-time digital transactions, as well as exceeding the global daily volume of Visa at 640 million daily transactions (NPCI, 2026). New monthly peaks are being breached regularly: in August 2025, there were 20+ billion transactions; December 2025 set a peak of 21.63 billion; in January 2026, the peak was 21.70 billion; and in April 2026, a peak of 22.35 billion was established (NPCI, May 2026). The current success ratio of UPI is at 99.2%, where 684 banks are live, and it has a total of 491 million users. Only person-to-merchant (P2M) transactions accounted for a figure of 67.01 billion in H1 2025, up 37% from last year. By FY2026-27, UPI transactions are projected to reach 379 billion annually, constituting 90% of all retail digital payments (PwC India, 2026).

Metric	FY2021	FY2022	FY2023	FY2024	FY2025	Apr 2026 (Latest)
UPI Vol. (Billion)	22.3	46.0	83.7	131.0	250+	22.35 (monthly)
UPI Value (USD Tn)	0.41	0.89	1.52	2.22	3.4	
Active Merchants on PGs (M)	3.5	4.8	6.2	9.0	12.0+	12M+
India FinTech Mkt (USD Bn)	50	73	111	142.5	142.5	USD 595B by 2034
UPI Banks Live	223	304	365	509	680+	684
UPI Active Users (M)			300	350	491	491

Table 1.1 *India Digital Payments Key Metrics 2021–April 2026 (Sources: NPCI 2026; IMARC Group 2026; RBI; PwC India 2026)*

1.3 Problem Statement

While there are plenty of studies done by practitioners on fintech disruptions and increased academic attention towards platform economics, specific gaps continue to exist in terms of analysis: the academic and managerial studies on the application of API technology by B2B payment infrastructure firms moving from single-product payment acceptance to an integrated, AI-driven financial platform is sparse, especially for an emerging market context. What is the impact on the competition within the payments ecosystem if the addition of agentic AI systems changes the game from advice-giving to action-taking? How does integration via Model Context Protocol (MCP) Server change the developer switching cost equation?

Razorpay, The leading B2B payment infrastructure firm in India is the best example of a theoretically and empirically sophisticated case study to study such issues. Founded in 2014, Razorpay has matured into a full-stack financial services firm that serves more than 12 million businesses (Q1 2026) and processes about one billion transactions per quarter with a TPV of approximately USD 45 billion per quarter (annualized: USD 180 billion) to become a player holding more than 55% share of India's online payment gateway market. Razorpay is the first payments firm in the world to embed UPI and all Indian payment methods into OpenAI's Codex through integration with an MCP server in April 2026. Its Agent Studio launched in March 2026 based on Anthropic's Claude Agent SDK uses AI-powered bots to handle disputes, payment recovery, and cash flow analysis. Razorpay filed a confidential IPO with SEBI in April 2026, aiming to raise USD 600-700 million with an enterprise value of USD 5-6 billion. However, no academic study has explored this journey of Razorpay so far.

1.4 Research Objectives

This study is guided by five primary research objectives:

1. To examine the macro-structural forces technological, regulatory, economic, and social driving innovation in financial services globally and within India between 2014 and May 2026.
2. To analyse Razorpay's product architecture, business model, and competitive strategy, including the 2025–2026 additions of Agent Studio, OpenAI Codex integration, and international expansion as a case study in platform-based financial services innovation.
3. To evaluate impacts on merchant efficiency, credit access, settlement visibility, retention, liquidity stability, and conditional growth.
4. To develop and update the DELTA Model of FinTech Competitive Advantage, extending it with Stage A+ (Agentic Intelligence) to reflect the qualitative shift from passive infrastructure to active financial intelligence.
5. To derive actionable, evidence-grounded strategic recommendations for incumbent financial institutions, fintech firms, SME users, and regulatory policy makers updated for May 2026 context including the anticipated IPO, UPI's 25-country expansion, and agentic AI governance requirements.

1.5 Research Questions

- RQ1: What structural forces have driven financial services innovation in India between 2014 and May 2026, and how do these compare with global fintech trajectories?
- RQ2: How has Razorpay's API-first architecture and platform strategy generated competitive advantages that differ qualitatively from traditional financial institution strategies and how does the addition of agentic AI alter this competitive architecture?

• RQ3: What is the impact of Razorpay's embedded financial solutions, such as Razorpay Capital with its quick disbursal facility and Line of Credit up to Rs. 25 lakh, on credit access by SMEs, and what is the significance of this phenomenon in terms of financial inclusion theory via digital platforms?

• RQ4: Can an extended DELTA Model (including Stage A + Agentic Intelligence) help explain the trajectory of Razorpay since its inception in 2014 to 2026, and what are some generalizable lessons that emerge through this process?

• RQ5: What are some potential strategies that incumbents, fintech start-ups, users, and regulators can undertake in light of the evolving nature of the industry due to the advent of agentic intelligence and API-based platforms?

1.6 Scope and Limitations

1.6.1 Scope

The time frame ranges from the formation of the company in 2014 until May 2026, which marks the latest date for which data is available at the time of writing this updated report. The geographical scope is mostly limited to India, but comparisons will be drawn with other countries' fintech industries including USA, UK, China, Brazil, Malaysia, and Middle East. The scope of the industry includes payments, banking, lending, payroll, compliance, and AI-based finance operations.

1.6.2 Limitations

- Data Constraint: Razorpay remains a private company without published audited financial statements as of May 2026. FY25 revenue of Rs 3,783 crore (65% YoY growth) and net loss of Rs 1,209 crore are derived from Registrar of Companies filings. The IPO DRHP when publicly filed with SEBI following the April 2026 confidential filing will materially improve data availability.

• IPO Timing Uncertainty: Razorpay's confidential IPO filing with SEBI (April 2026), seeking funding between $600-700 million against a valuation of $5-6 billion, has not yet seen the release of its public DRHP. The IPO result is considered as forward-looking.

• Regulatory Currency: The Digital Lending Directions 2025 replaces the earlier version issued in 2022; while the Data Privacy and Protection Act 2025 replaces the one passed in 2023. All regulatory sources are as per May 2026.

• Agentic AI Novelties: The Agent Studio (March 2026) and OpenAI Codex (April 2026) developments have happened too recently to be subject to academic research scrutiny.

1.7 Significance of the Study

The significance of this research is multi-dimensional. For academic scholars, it advances the literature on platform economics and disruptive innovation theory by applying them to an emerging-market fintech context that is both understudied and, as of May 2026, the world's most active real-time payments laboratory. For practitioners, it provides the updated DELTA Model extended with Agentic Intelligence as a structured analytical framework for understanding competitive dynamics in AI-driven financial services markets. For policy makers, it surfaces empirical evidence that embedded finance with AI-driven underwriting can improve credit targeting, settlement visibility, and risk screening. For MBA students and researchers, it provides a methodologically rigorous template applicable to other platform-based industries undergoing AI-driven transformation.

1.8 Report Structure

This report is structured as follows: Chapter 2 reviews academic and practitioner literature across seven thematic domains (adding Agentic AI as a new seventh domain in this 2026 edition). Chapter 3 details the research methodology. Chapter 4 presents the industry and company analysis, updated through May 2026. Chapter 5 analyses data and performance insights. Chapter 6 examines AI/ML models and Agent Studio implications. Chapter 7 discusses findings. Chapter 8 introduces and extends the DELTA Model. Chapter 9 presents strategic recommendations updated for 2026–27. Chapter 10 concludes the dissertation. Three appendices cover Power BI dashboards, AI/ML model design, and data reference.

CHAPTER 2

Review of Literature

This chapter synthesizes academic and practitioner research across seven thematic domains that collectively form the theoretical foundation for this dissertation. The review spans literature from 2000 to May 2026, with particular attention to developments in the 2022–2026 period that substantially reshape established frameworks. For each domain, the review presents the foundational scholarship, identifies consensus positions, notes contradictions, and highlights gaps that this study addresses. A new seventh domain Agentic AI in Financial Services is introduced in this 2026 edition, reflecting the material significance of autonomous AI systems to Razorpay's 2025–2026 competitive trajectory.

2.1 Financial Innovation Theoretical Foundations

2.1.1 The Schumpeterian Tradition

Joseph Schumpeter's concept of 'creative destruction,' articulated in Capitalism, Socialism and Democracy (1942), provides the bedrock theoretical context for understanding how fintech platforms like Razorpay challenge established financial institutions. Schumpeter argued that capitalist progress advances not through smooth, marginal improvements to existing methods, but through periodic, disruptive reorganizations innovations that simultaneously destroy the commercial viability of existing production functions while creating superior ones. The term 'creative destruction' captures this duality: new firms, methods, and products rise precisely by making incumbent arrangements economically untenable.

Applied to financial services, this thesis predicts with remarkable accuracy the dynamics of the 2014–2026 period. Razorpay did not compete against existing Indian payment gateways on their own terms; it made the prior generation's integration complexity commercially unjustifiable by offering a developer-friendly API that reduced integration time from weeks to hours, pricing transparency that eliminated opaque fee structures, and digital merchant verification that removed the six-week physical onboarding requirement of legacy processors. Subsequent Razorpay products RazorpayX, Neobanking, Razorpay Capital lending, and the March 2026 Agent Studio have followed the same creative-destructive logic, each occupying a function that incumbent banks theoretically perform but at a speed, cost, and integration depth that the incumbent branch-centric model structurally cannot match.

Critical evaluation: The Schumpeterian framework has important limitations in financial services specifically. Financial regulation creates substantial friction that slows the revolutionary dynamic, for two reasons. First, licensing requirements, capital norms, and consumer protection legislation favour incumbents who have already made the compliance investment   the very compliance infrastructure that makes innovation entry costly simultaneously protects the innovator's position once achieved. Second, the systemic importance of financial institutions to the broader economy gives regulators a legitimate interest in constraining the pace of disruption that has no equivalent in other industries. This regulatory friction explains why, despite the fintech revolution of the 2010s and 2020s, incumbent banks have not lost significant deposit or loan book market share in most jurisdictions despite Schumpeterian predictions to the contrary. The regulatory friction analysis developed by Zetzsche et al. (2017) provides the necessary corrective to the pure Schumpeterian framework.

A 2026 extension of the Schumpeterian argument is required. Autonomous AI agents such as those deployed through Razorpay's Agent Studio (March 2026, built on Anthropic's Claude Agent SDK)   that independently initiate, execute, and reconcile financial operations represent a new form of creative destruction. This is not the destruction of a product category or a distribution channel, as in earlier fintech waves, but the destruction of the operational labour model within financial services itself. Davidovic and Tourpe's IMF Note (2026) is among the first institutional publications to formalise this argument, observing that agentic AI systems can 'shift transactions from human-initiated instructions to agent-mediated decisions'   a Schumpeterian reorganisation of the human-machine division of financial labour that has no historical precedent.

2.1.2 Disruptive Innovation Theory

Christensen's Innovator's Dilemma (1997) provides a more granular and operationally useful model of fintech competitive dynamics than Schumpeter's macro-historical framework. Christensen distinguished sustaining innovations, improvements to existing products along dimensions that mainstream customers already value from disruptive innovations, which initially appear inferior on established performance metrics but excel on dimensions that incumbents have previously dismissed as commercially unimportant: simplicity, accessibility, affordability, and convenience for users who have been overserved or entirely excluded from existing solutions.

The Indian fintech context maps particularly cleanly onto Christensen's non-consumer disruption variant. The version of his model with the strongest empirical record. The initial target market for Razorpay was not the large enterprise merchants served by established gateways, but the 63 million MSMEs excluded from digital payments entirely by integration complexity and the gig-economy platforms and early-stage startups that existing processors dismissed as commercially marginal. By solving problems that incumbents regarded as unimportant, Razorpay built scale on a customer base that incumbents were not actively defending. The subsequent 'upmarket march' Christensen's prediction that disruptors progressively improve their products to compete for mainstream customers is empirically visible: from serving early-stage startups in 2014 to processing payments for IRCTC, Airtel, Zomato, and Meesho by 2024–2026, and from basic payment gateway services to Agent Studio agentic AI operations by 2026.

The broader academic evidence base for Christensen's model has been contested. King and Baatartogtok (2015) re-examined 77 cases that Christensen himself classified as disruptive and found that fewer than 10% fully followed the predicted trajectory. However, this critique applies most forcefully to Christensen's market-segmentation disruption variant; the non-consumer variant which describes Razorpay's founding logic has a substantially stronger predictive record and a more rigorous theoretical grounding in genuine market access barriers rather than customer preference dynamics.

Christensen et al. (2015) later clarified that not all competitive threats to incumbents qualify as disruption in the technical sense. By this refined definition, Razorpay's original payment gateway displacing the unserved market, not competing feature-for-feature with established gateways qualifies fully. The Agent Studio, by contrast, represents operational disruption rather than market disruption: it targets the human-managed financial operations of merchants already using Razorpay, not a new non-consuming market. This distinction carries strategic implications for how the competitive threat from agentic AI should be characterized in incumbent bank strategy.

2.1.3 Frame and White's Innovation Taxonomy

Frame and White (2004) proposed a three-part taxonomy of financial innovation that remains analytically valuable because it reveals the structural character of fintech's competitive advantage: process innovations (new delivery mechanisms for existing financial services), product innovations (new financial instruments or contracts), and organisational innovations (new ownership structures, distribution arrangements, or institutional designs). Their central empirical finding that financial innovation is overwhelmingly processual and organisational rather than genuinely new in product terms explains why competitive differentiation in digital payments is so operationally intensive.

Every major Razorpay innovation is classifiable as either process or organisational: the API payment gateway is a process innovation; RazorpayX is an organisational innovation distributing banking products through a non-bank platform; Razorpay Capital is an organisational innovation distributing credit through the same platform; and the Agent Studio is a process innovation of a qualitatively new kind one that does not redesign the process for human execution but eliminates human execution from the process entirely.

This classification carries an important competitive implication that Frame and White's original framework did not fully develop: processual and organisational innovations are more replicable than product innovations. A new financial instrument requires deep technical and legal expertise that is genuinely difficult to copy; a new process or distribution arrangement can, in principle, be matched by any competitor with sufficient engineering resources and regulatory standing. This replicability dynamic explains both the intensity of competitive investment in developer experience and API quality across Indian payment platforms, and the significance of Agent Studio as a new competitive dimension one where the operational intelligence accumulated through AI training on millions of merchant interactions is more difficult to replicate than the underlying agentic architecture itself.

This study proposes extending Frame and White's taxonomy with two additional categories appropriate for the 2025–2026 context: data-intelligence innovations, which use accumulated transaction and behavioural data to create underwriting, personalisation, and prediction capabilities with no precedent in traditional finance; and operational autonomy innovations, which transfer the execution of human-managed financial processes to autonomous AI agents capable of reasoning and action within defined operational guardrails. Both types are empirically instantiated in Razorpay's 2014–2026 product trajectory.

2.1.4 Agentic AI as a New Innovation Category (2026)

The rapid deployment of agentic AI systems in financial services during 2025–2026 characterised by systems that can 'interpret objectives, break them into tasks, and interact with digital services with limited human input' (Davidovic & Tourpe, IMF, 2026) represents a fifth category of financial innovation that neither Frame and White nor any prior classification anticipated. The distinction from process automation is qualitative, not incremental. Process automation redesigns financial workflows for more efficient human execution; operational autonomy innovation removes human execution from financial workflows entirely, replacing it with AI agents that observe operational signals, reason over structured data, and take consequential actions within compliance-defined guardrails.

This is clearly shown by the financial services industry data for the structural change occurring in 2025-2026 and not just a trial phase of the pilot programme. The adoption of AI technology within financial services increased from 37% of companies using it in 2023 to 59% in 2025, while the market size of the autonomous agent market in the financial sector is expected to rise from USD 7.8 billion in 2025 to USD 52 billion in 2030, while financial services firms have been employing autonomous agents in areas like fraud detection (64%), loan processing (61%), and customer onboarding (59%) (PiTech Solutions, 2026).

2.2 Platform Economics and Two-Sided Markets

2.2.1 Rochet-Tirole Two-Sided Market Theory

Rochet and Tirole's (2003) formal analysis of two-sided market economics provides the theoretical framework most directly applicable to understanding payment gateway competitive dynamics. A two-sided platform intermediates transactions between two distinct user groups whose value from the platform is interdependent: merchants require a critical mass of consumer payment methods accepted; consumers through their payment app choices require those apps to be accepted by a critical mass of merchants. This interdependence generates network effects whose logic produces winner-takes-most outcomes when network externalities are strong and switching costs are non-trivial.

The practical implication for Razorpay is direct. Its early strategy of maximizing supported payment methods (100+) while minimizing integration friction simultaneously expanded merchant-side value every new payment method accepted meant fewer lost transactions and consumer-side coverage every new merchant accepting Razorpay-processed payments meant the consumer's preferred payment app was usable in one more commercial context. The resulting network effects have proven durable: Razorpay now commands approximately 55% of India's online payment gateway market, a concentration precisely consistent with the winner-takes-most prediction of two-sided market theory.

Rochet and Tirole's model also yields a normative prediction about optimal platform pricing: the platform should subsidies the side with the lower willingness to pay or greater price sensitivity while charging the side with stronger demand for the other side's participation. India's UPI policy zero consumer charges, differential merchant settlement is precisely this optimal structure at the infrastructure level. Razorpay's decision to absorb the zero-MDR mandate on UPI transactions while generating revenue from higher-margin banking and lending products at the platform level mirrors the same pricing logic applied one layer up.

2.2.2 Multi-Sided Platforms and Developer Ecosystem Dynamics

Parker and Van Alstyne (2005) extended the two-sided market framework to platforms with more than two user groups, identifying developer ecosystems as a critical third side whose participation determines long-run platform competitiveness. Their central insight that platforms do not merely compete on product features or price but on their ability to attract developers who build complementary applications deepening the platform's value to all user groups is empirically central to understanding Razorpay's competitive durability.

Razorpay's investment in developer experience eight-language SDK coverage, an interactive API playground, comprehensive documentation that consistently receives developer community recognition, and Razorpay Labs is explicitly a platform competition strategy. Each of the 30+ e-commerce integrations (WooCommerce, Shopify, Magento) built by third-party developers expands Razorpay's addressable merchant universe without proportional investment. More importantly, each merchant who has built operational workflows using Razorpay's data models, webhook architecture, and API endpoints has incurred switching costs that scale with the sophistication of their integration rendering rational migration progressively less attractive as integration depth increases.

Boudreau and Lakhani (2009) examined the governance trade-off in platform openness: greater openness accelerates ecosystem growth and third-party innovation but reduces platform control over quality, competitive dynamics, and data access. Razorpay's approach resolves this trade-off through selective openness payment acceptance and payout APIs are fully open and well-documented for third-party integration; core financial ledger operations, credit underwriting models, and Agent Studio AI infrastructure are closed and controlled. This is the optimal resolution identified by Boudreau and Lakhani: maximise ecosystem density at the integration layer while protecting the data intelligence and operational capabilities that constitute competitive moat.

2.2.3 MCP Servers as Developer-Side Lock-in (2026)

A development of particular significance for platform economics research is Razorpay's April 2026 integration of its payment infrastructure into OpenAI's Codex platform via a Model Context Protocol (MCP) Server. The Model Context Protocol an open standard released by Anthropic in 2024 defines how AI models interact with external tools, APIs, and data sources, creating a standardised interface between large language models and the software infrastructure of the external world.

By deploying an MCP Server exposing Razorpay's payment APIs to AI coding agents, Razorpay has created a new mechanism for developer-side network effects that operates at a higher abstraction level than traditional SDK integration. Codex, processing over one million developer sessions weekly, can now embed UPI and all Indian payment methods into any new application through natural-language description collapsing integration time from days to under five minutes (Business Standard, April 2026). The competitive implication, which has received no prior academic treatment, is that MCP-mediated switching costs operate differently from SDK-level switching costs: rather than the cost of rewriting code to use a competing gateway's API, the switching cost involves re-prompting or re-training the AI agent that has been optimised for Razorpay's MCP interface. As AI-mediated software development scales, this mechanism may prove more durable than any previous developer-side competitive moat.

2.3 Digital Payments and the UPI Ecosystem

2.3.1 Theoretical Foundations of Network-Based Payment Design

The academic literature on payment systems economics intersects network theory, information economics, and regulatory design. Rochet and Tirole's interchange fee analysis (2003) provides the theoretical underpinning for India's zero-MDR UPI policy: eliminating consumer-side transaction cost to maximise network participation produces superior social welfare outcomes compared with any pricing scheme that imposes costs on either side. The empirical validation of this theory in India's context is uniquely powerful: UPI's 22.35 billion monthly transactions in April 2026, representing 50% of global real-time digital transaction volume (NPCI, 2026), provides a quasi-experimental demonstration that public-infrastructure payment design at scale outperforms private network economics in adoption breadth.

Halan and Sane (2019) provided systematic empirical documentation of India's digital payments adoption barriers circa 2018: high MDR rates, network unreliability, complex eKYC documentation, integration complexity, and cash-cultural inertia among merchant communities. Their five-barrier framework remains analytically important in 2026 because it establishes the counterfactual baseline from which the subsequent resolution of each barrier through UPI's zero-MDR mandate, Aadhaar-based digital KYC, API-first gateways, COVID-19-accelerated behavioural change, and progressive merchant QR code penetration produced the extraordinary growth trajectory documented throughout this dissertation.

2.3.2 UPI's Architecture and the 2016–2026 Growth Trajectory

UPI's structural design reflects a deliberate policy commitment to payment infrastructure as a public good: real-time, interoperable, zero-cost for consumers, and accessible to any licensed Payment Service Provider through open API architecture. This design philosophy differs fundamentally from the private, proprietary models dominant in the United States (Visa, Mastercard interchange economics) and China (Alipay and WeChat Pay ecosystem lock-in), and the comparative outcomes are instructive: UPI's active user base of 491 million including hundreds of millions previously entirely outside formal payment systems is proportionally the highest financial inclusion outcome of any national payment system in history.

As confirmed by the Payments Council of India, UPI accounts for over 84% of all digital payments made in India on a volume basis. As a result of the maturity of the underlying infrastructure for IMPS as well as optimization algorithms powered by ML being used by participant banks and payment aggregators such as Razorpay, transaction success stands at 99.2% by 2026. Person to Merchant transactions stood at 67.01 billion for H1 of 2025, which represents a YOY growth of 37% and is aided by 678 million merchant QR touchpoints.

2.3.3 UPI's 2025–2026 Milestones and International Expansion

UPI statistics for the most recent period, as of the date of this writing (NPCI, May 2026), show a further acceleration trend without any signs of market saturation. April 2026 witnessed a record number of 22.35 billion transactions after a series of previous highs: August 2025 (20+ billion, first time surpassing this level), December 2025 (21.63 billion high), January 2026 (21.70 billion), then a seasonal drop to February 2026 (20.39 billion), followed by recovery in April 2026. Year-over-year, the UPI system will likely hit about

268 billion transactions in 2026, while PwC India (2026) forecasts 379 billion annually by FY2026-27, or 90% of all retail payments digitally.

In parallel, India has been exporting the UPI system to 25 countries by April 2026, including Israel, with the latter being included in the latest expansion during PM Modi's state visit to the country. The Project Nexus initiative connecting the fast payment systems of India, Malaysia, Thailand, the Philippines, Singapore, and Indonesia is anticipated to form an integrated instant payment corridor between the six countries for more than 1.7 billion people in 2026. Such international innovations serve as evidence of the fact that analyzing India's payment system innovations is highly relevant on a global scale.

Period	Monthly Volume (Bn)	Annualised Run Rate (Bn)	Key Milestone
FY2024 Full Year	10.9 avg	131	Surpassed Brazil as largest real-time market
August 2025	20.0+	240+	First month crossing 20 billion threshold
October 2025	16.6	199	UPI Autopay: 25% YoY growth milestone
December 2025	21.6	259	Calendar year peak festive + year-end
January 2026	21.7	260	New monthly record at time
February 2026	20.4	245	Seasonal moderation (expected)
April 2026	22.35	268	Latest record 50% of global real-time volume

Table 2.3 *UPI Monthly Transaction Volume: August 2025–April 2026 (Source: NPCI, May 2026)*

2.4 Embedded Finance and the API Economy

2.4.1 Defining Embedded Finance

Embedded finance describes the integration of financial products lending, insurance, banking, investment, payments into non-financial digital platforms such that users access those services within familiar operational contexts rather than navigating to dedicated financial institutions. The concept extends open banking: while PSD2-style open banking focuses on regulated data sharing between licensed banks and authorised third parties, embedded finance goes further by enabling any sufficiently data-rich digital platform with a regulated backend partner to become the primary financial services distribution interface for its users.

Mention and Torkkeli's (2012) service-dominant logic framework provides useful theoretical grounding: in service-dominant logic, value is not delivered unilaterally by institutions but co-created through platform-customer interaction and data exchange. Embedded finance is the most complete operational expression of this logic financial products are explicitly shaped by the customer's operational context, surfaced within workflows the customer already uses daily, and priced using data that the customer has already generated through their platform activity. The contrast with institution-centric financial services where the customer must navigate an institution's product framework rather than receiving products adapted to their operational reality captures precisely the informational and experiential advantage that embedded finance creates.

McKinsey's (2023) market sizing of the global embedded finance opportunity at USD 7 trillion by 2030 has been widely cited. More recent estimates from Fortune Business Insights (2026) place the global embedded finance market at USD 148.38 billion in 2025, growing toward USD 588 billion by 2033 at an 18.5% CAGR. While methodology differences across estimates are material, the directional consensus is unambiguous: embedded finance has transitioned from a niche fintech experiment to a structural redesign of financial services distribution.

2.4.2 The API Economy and Developer-Led Growth

The 'API economy' thesis articulated academically by Greenstein and Nagle (2014) and validated commercially by the trajectories of Stripe, Plaid, and Razorpay holds that economic value creation increasingly organizes itself around the exchange of services through programmable interfaces. The significance for competitive strategy is that API quality becomes the primary determinant of ecosystem density, and ecosystem density becomes the primary determinant of competitive durability in platform-mediated industries.

Three mechanisms by which API quality creates competitive advantage are identified in the literature: network effects (each additional developer integration expands addressable merchant universe); switching costs (merchants who have built operational workflows using a platform's data models and webhooks incur restructuring costs to migrate); and complementor ecosystems (third-party applications accounting software, inventory systems, analytics tools built on a payment API create additional reasons for merchants to remain on the platform). Together, these constitute the 'API moat' whose depth is the primary long-run determinant of competitive durability in payment infrastructure.

The 2026 addition to this framework, as discussed in Section 2.2.3, is the MCP Server as a fourth mechanism of API-economy competitive advantage: AI-mediated integration. As AI coding agents progressively handle payment infrastructure implementation, the competitive advantage of having the best-documented, most developer-accessible API compounds through the AI layer and the switching cost of changing payment providers becomes a question of AI model re-optimisation, not merely code migration.

2.4.3 India's Embedded Finance Market: 2025–2026 Update

As per figures from Research and Markets in 2025, the total size of India’s embedded finance market stood at \$24.03 billion – reflecting a YoY growth rate of 12.4% and registering a CAGR of 17.8%. The segment-wise market breakup of India’s embedded finance market includes digital lending (approximately 43%); embedded payments (34%); embedded insurance (13%) and embedded investments (10%). In addition, India's alternate lending market crossed the \$30.57 billion mark – growing at a CAGR of 14.3%, with fintech platforms providing an increasing amount of credit to MSMEs.

Competitive structure is significantly evolved compared to 2023 levels. Platform-based distribution has become standard practice in the field of SME digital finance in India. One example of this is Razorpay Capital, where the firm offers 10-second disbursals along with Line of Credit up to Rs 25 lakh at a monthly interest rate of 1.5% without requiring any collateral. B2B segment. The backend has consolidated around a small number of BaaS enablers and regulated NBFC partners, while front-end interfaces remain fragmented. Critically, platforms with more granular, real-time transaction data like Razorpay systematically achieve lower credit loss rates than those relying on traditional bureau-based underwriting, providing empirical support for the data-first underwriting thesis at the foundation of the DELTA Model.

2.4.4 The SME Credit Gap and Embedded Lending Economics

The MSME finance gap in India stands at USD 530 billion, according to the World Bank (2022). It is defined as the difference between the aggregate need for MSME formal and informal financing minus the amount formally financed by financial organizations. The conservative estimate of unmet MSME financing demand stands at USD 240-300 billion by Markntel Advisors (2026), highlighting the fact that less than 20% of MSME finance demands have been catered to formally. Both estimates highlight the failure of the market based on the problem of information asymmetry.

The economics of embedded lending differ from traditional NBFC or bank lending in a way that is analytically fundamental. A platform that has processed 12 months of a merchant's payment transactions has a real-time, comprehensive picture of the merchant's revenue, seasonality, cash conversion cycle, customer concentration, and payment behaviour a dataset that is both more current and more granular than the periodic financial statements and collateral valuations that traditional underwriters use. This data advantage enables two measurable superiorities: higher approval rates for creditworthy merchants traditionally declined on documentation grounds, and lower default rates because the continuous data feed enables real-time early-warning signals that periodic statement review cannot provide.

2.5 FinTech Regulation and Risk

2.5.1 The Regulatory Trilemma

According to Zetzsche et al. (2017), the key difficulty in the governance of fintech is what the authors define as a ‘regulatory trilemma’. This involves reconciling the need for maintaining financial stability, consumer protection, and fostering innovation, which proves to be hard since there are inherent conflicts between the goals. Too restrictive policies on stability create hurdles to innovation. Likewise, a higher degree of consumer protection raises the compliance costs and thus is costlier for new market players than established companies.

India's resolution of this trilemma has been extensively noted as more successful than comparable economies enabling fintech innovation at extraordinary scale while maintaining systemic financial stability. The RBI's 'calibrated risk-taking' approach enabling innovation through framework development and regulatory sandboxes rather than blanket restriction, while maintaining systemic integrity through progressive licensing requirements and conduct rules has produced an environment where fintech growth and regulatory coherence have co-evolved rather than conflicted.

The 2026 extension of Zetzsche et al.'s trilemma introduces a fourth dimension: AI governance. Autonomous AI agents conducting financial transactions introduce risks that existing frameworks do not comprehensively address accountability (who is liable when an AI agent makes an erroneous dispute recovery decision?), transparency (are merchants adequately informed that their financial operations are managed by an AI?), and systemic risk (do correlated AI agent behaviours across multiple platforms amplify financial stress There have been five regulatory events that have had a significant impact on Razorpay’s environment until May 2026. The Payment Aggregators Guidelines (RBI, 2020) introduced the first regulatory framework for payment aggregators, requiring all firms handling settlements for merchants to be authorised by RBI, have adequate capital, follow data protection norms, and have ring-fenced customer fund escrow accounts. The Account Aggregator Framework (RBI, 2021) introduced the structure for sharing financial data consent-based that facilitates multi-source underwriting without having to share documents physically.

events?). The IMF Note by Davidovic and Tourpe (2026) provides the most authoritative current treatment, but its framework remains preliminary and regulatory responses remain unsettled globally.

2.5.2 India's Regulatory Milestones Through May 2026

The Digital Lending Guidelines (RBI, 2022) and Digital Lending Directions (RBI, 2025) between regulated lenders and borrowers, explicit FLDG disclosure, and standardised fee and interest rate disclosure. The Digital Personal Data Protection Act (DPDP Act, 2025), superseding the 2023 version with strengthened enforcement provisions, creates material obligations for Razorpay as a large-scale processor of merchant and consumer financial data. The RBI's 2025 cross-border Payment Aggregator approval enabled Razorpay to process international payments a prerequisite for the Stage A international expansion documented in the DELTA Model.

2.6 Agentic AI in Financial Services   New Domain (2026)

2.6.1 Theoretical Framework

Agentic AI systems   defined by Davidovic and Tourpe (IMF, 2026) as AI systems capable of 'interpreting objectives, breaking them into tasks, and interacting with digital services with limited human input'   represent the most consequential development in financial services technology since the smartphone. The distinction from prior AI applications in finance (ML-based fraud detection, NLP chatbots) is qualitative: those systems advised human operators who retained decision authority; agentic systems act, executing consequential financial operations autonomously within defined guardrails.

Acharya et al. (2025) identify four architectural components of effective agentic AI systems in financial compliance: a large language model providing reasoning capability; tool-use integrations enabling the agent to interact with external APIs, databases, and financial systems; guardrails implementing regulatory and operational constraints that bound agent actions; and escalation protocols routing complex or uncertain cases to human oversight. This architecture maps directly onto Razorpay's Agent Studio design (Anthropic Claude SDK providing LLM reasoning; Razorpay APIs providing tool-use integrations; compliance-defined guardrails; human escalation for high-value cases).

The competitive significance of agentic AI deployment in payment infrastructure operates through two mechanisms not present in any prior fintech competitive dynamic. First, AI model calibration switching costs: Razorpay's Dispute Responder agent has been trained on the specific evidence patterns, dispute taxonomies, and resolution pathways relevant to Razorpay's specific payment methods and merchant base. A merchant who switches payment providers loses this accumulated agent intelligence   a switching cost that scales with the duration and depth of agent deployment. Second, operational intelligence accumulation: as agents process more disputes, recover more payments, and forecast more cash flows, their outputs improve continuously creating a performance advantage over new entrant competitors whose agents have not yet accumulated comparable operational experience.

2.6.2 Empirical Evidence from 2025–2026 Deployments

The global use of agentic artificial intelligence (AI) in the financial sector increased significantly in 2025. According to CB Insights (2026), all major payment processing companies such as Mastercard, Visa, Stripe, and Shopify implemented agentic commerce in 2025. The agentic AI market in the financial services industry is forecasted to increase from $7.8 billion in 2025 to $52 billion by 2030 (PiTech Solutions, 2026). Autonomous agents were mainly implemented by financial institutions in fraud detection (64%), loan processing (61%), and customer onboarding (59%).

In India, Razorpay's Agent Studio represents the most advanced published case of agentic deployment in payment infrastructure at the time of writing. Three production agents launched March 2026: Dispute Responder (autonomous chargeback evidence generation), Failed Payment Recovery (AI-driven retry workflow management), and Cash-Flow Forecasting (autonomous liquidity scenario modelling). A No-Code Agent Builder in beta extends autonomous agent creation to finance teams without engineering resources. The World Economic Forum (2026) has begun developing 'Know Your Agent' verification protocols for agentic commerce, analogous to KYC for human transactors recognition that the governance infrastructure for AI agents in financial services is being built in real time in response to production deployments like Agent Studio.

2.7 Research Gap and Theoretical Contribution

The review of literature across seven thematic domains reveals four research gaps that this dissertation addresses:

Gap 1: Application to India's agentic-AI-enhanced fintech ecosystem: The academic literature on disruptive innovation theory and platform economics is rich but has been applied almost exclusively to US and European fintech cases. Razorpay's case embedded in India's UPI infrastructure, operating under India's specific regulatory architecture, and now deploying agentic AI at the global frontier of payment innovation offers a distinct analytical context that tests the generalisability of Western-derived theoretical frameworks.

Gap 2: Agentic AI as competitive advantage mechanism: No existing fintech competitive advantage framework including earlier versions of the DELTA Model was developed with agentic AI as a variable. The Stage A+ extension proposed in Chapter 8 is the first attempt in academic literature to incorporate autonomous AI agents as a distinct and theoretically grounded source of fintech competitive advantage.

Gap 3: MCP Server integrations as platform lock-in: The Model Context Protocol, deployed commercially by Razorpay in April 2026, has received no prior academic analysis. Its significance for platform competition creating AI-mediated switching costs at a higher abstraction level than SDK integration is identified as a priority for future platform economics research.

Gap 4: Multi-case DELTA Model validation: The DELTA Model (Chapter 8) is inductively derived from Razorpay's case and partially corroborated by comparison with Stripe, Nubank, and Alipay. Rigorous empirical validation requires a formal multi-case research design across API-first fintech platforms in diverse market contexts, representing the clearest direction for future research following from this dissertation.

Author(s)	Year	Core Contribution	Theme	Critical Evaluation
Schumpeter	1942	Creative destruction as driver of capitalist progress	Innovation Theory	Foundational; under-theorises regulatory friction; extended by Davidovic &

Author(s)	Year	Core Contribution	Theme	Critical Evaluation
				Tourpe (2026) for agentic AI
Christensen	1997	Disruptive innovation non-consumer variant	Strategy	Strong empirical record for non-consumer variant; contested for market-segmentation variant
Rochet & Tirole	2003	Two-sided platform economics; optimal interchange pricing	Platform Economics	Core framework for payment network strategy; directly applicable throughout
Frame & White	2004	Process/product/organisational innovation taxonomy	Innovation Taxonomy	Requires two additional types: data-intelligence and operational autonomy (2026 extension)
Parker & Van Alstyne	2005	Multi-sided platforms and developer ecosystem strategy	Platform Strategy	Applicable to Razorpay API and developer community investment rationale
Boudreau & Lakhani	2009	Platform openness governance trade-off	Platform Governance	Provides basis for Razorpay's selective openness strategy
Mention & Torkkeli	2012	Service-dominant logic in financial innovation	Embedded Finance Theory	Grounds embedded finance in value co-creation rather than institutional delivery
Greenstein & Nagle	2014	API economy and programmable interface value	API Economy	Academic basis for API moat thesis; MCP extends to AI abstraction layer
Arner, Barberis, Buckley	2016	FinTech 3.0 disaggregation of financial services	FinTech History	Most-cited fintech history framework; India context under-developed

Author(s)	Year	Core Contribution	Theme	Critical Evaluation
Halan & Sane	2019	Five-barrier model of India digital payments adoption	India Payments	Empirically rigorous; provides counterfactual baseline for UPI growth drivers
Thakor	2020	Fintech-banking intersection; systemic risk of algorithmic credit	Banking & FinTech	Systemic risk concerns newly relevant for agentic AI operations
Zetzsche et al.	2017	Regulatory trilemma: stability vs protection vs innovation	FinTech Regulation	Best policy framework available; extended with AI governance as fourth dimension
McKinsey	2023	Global embedded finance market opportunity USD 7 trillion	Embedded Finance	Influential market sizing; superseded by 2025–2026 estimates in specific projections
NPCI	2026	UPI statistics: 22.35 billion monthly transactions, Apr 2026	India Payments Data	Primary source; confirms 50% of global real-time volume
IMF (Davidovic & Tourpe)	2026	Agentic AI reshaping payments governance	Agentic AI	First authoritative institutional framework for AI agent governance in payments
Fortune Business Insights	2026	Global fintech USD 460.76 billion (2026), 18.2% CAGR	Market Data	Most current global fintech market sizing
CB Insights	2026	Agentic AI deployment across financial services 2025–2026	Industry Data	Quantifies production AI agent adoption in financial services organisations
Research & Markets	2025	India embedded finance USD 24.03 billion (2025)	India Embedded Finance	Most current India-specific embedded finance market data

Author(s)	Year	Core Contribution	Theme	Critical Evaluation
World Economic Forum	2026	Know Your Agent governance protocols	Agentic AI Governance	Identifies regulatory gaps; confirms need for AI agent governance frameworks
RBI	2025	Digital Lending Directions 2025; Cross-border PA approval	India Regulation	Most current regulatory baseline for Razorpay Capital and international operations

Table 2.1 *Literature Review Matrix 20 Key Sources (Updated May 2026)*

CHAPTER 3

Research Methodology

3.1 Introduction

Research methodology is the systematic framework that governs how a study is designed, how data is collected and analysed, and how conclusions are drawn and validated. A clearly articulated methodology is not a bureaucratic formality but a scientific accountability mechanism: it allows readers, reviewers, and future researchers to evaluate the quality of the evidence on which the study's conclusions rest, to identify where methodological choices may have introduced bias or limitation, and to assess whether the findings are likely to generalise beyond the specific case studied.

This dissertation investigates financial services innovation in India through a case study of Razorpay, incorporating the most recent data and developments through May 2026. The nature of the research question what mechanisms explain the competitive dynamics of an API-first fintech platform evolving through agentic AI deployment? is both descriptive and explanatory in character, requiring a methodology capable of capturing both empirical patterns and theoretical mechanisms. This chapter details the research philosophy underpinning the design, the case study strategy, the data collection framework, the analytical methods applied, and the ethical considerations governing the research.

3.2 Research Philosophy

3.2.1 Philosophical Position

This research adopts a pragmatist philosophical position. Pragmatism holds that the most important criterion for evaluating a research approach is its practical utility its capacity to generate knowledge that illuminates real-world phenomena and produces actionable understanding rather than its conformity to a single ontological or epistemological tradition. This position is particularly appropriate for applied management research, where both the theoretical frameworks being tested (platform economics, disruptive innovation theory) and the empirical phenomena being explained (a specific company's competitive evolution) exist at the intersection of quantitative and qualitative evidence.

Pragmatism avoids the positivist-interpretivist dichotomy that constrains much management research methodology. Rather than insisting that either quantitative measurement or qualitative interpretation is the uniquely valid route to reliable knowledge, pragmatism holds that the research question itself should determine the method. For this dissertation, both modes of inquiry are necessary: quantitative analysis establishes the empirical scale and direction of Razorpay's competitive dynamics; qualitative analysis provides the mechanism-level understanding of how and why those dynamics operate as they do.

3.2.2 Research Design: Explanatory Mixed Methods

The research adopts Creswell and Plano Clark's (2018) explanatory sequential mixed-methods design. In this design, the qualitative strand is primary providing the theoretical and interpretive framework while the quantitative strand is secondary, providing empirical calibration, validation, and grounding for qualitative claims. The two strands are not parallel but integrated: quantitative findings raise questions that qualitative analysis answers, and qualitative frameworks identify which quantitative patterns are analytically significant.

The explanatory sequential design is appropriate for three reasons. First, the phenomena under investigation competitive advantage in platform-based financial services are inherently mechanism-oriented, requiring theoretical explanation rather than purely statistical description. Second, the data available from a private company (Razorpay has not published audited financial statements as of May 2026) is primarily secondary and estimated, making statistical inference in the probabilistic sense impossible; the quantitative strand supports analytical rather than inferential claims. Third, the theoretical contribution sought the DELTA Model and its Stage A+ extension is conceptual rather than statistical, requiring a methodology that privileges interpretive rigor over measurement precision.

3.3 Research Strategy: Explanatory Case Study

3.3.1 Justification for Case Study Approach

Yin (2018) defines a case study as an empirical inquiry that investigates a contemporary phenomenon within its real-world context, particularly when the boundaries between phenomenon and context are not clearly evident and when multiple sources of evidence are used. This definition describes precisely the methodological requirements of this dissertation: understanding Razorpay's competitive evolution requires understanding it within the specific context of India's UPI ecosystem, its regulatory architecture, and the 2025–2026 emergence of agentic AI as a competitive dimension each of which is inseparable from the phenomenon under study.

Yin (2018) identifies three types of case study design exploratory, descriptive, and explanatory distinguished by the nature of the research question. This dissertation is primarily explanatory (addressing 'how' and 'why' questions about Razorpay's competitive dynamics and the mechanisms that sustain them) with descriptive elements (documenting the empirical evolution of Razorpay's product suite and market position through May 2026). The hybrid character is theoretically appropriate: explaining a mechanism requires first accurately describing the phenomenon the mechanism is supposed to explain.

3.3.2 Single-Case Justification and Analytic Generalisation

A frequent methodological objection to case study research that single-case findings cannot be generalized reflects a category error about the type of generalisation case studies support. As Yin (2018) clarifies, case studies support analytic generalisation (from case to theory) rather than statistical generalisation (from sample to population). The DELTA Model proposed in Chapter 8 is analytically generalisable: it claims that the mechanisms it identifies data accumulation enabling embedded finance, embedded finance enabling lock-in, lock-in funding technology reinforcement, technology reinforcement enabling advancement, and agentic AI enabling operational intelligence moats operate in any API-first fintech platform that pursues a comparable strategic trajectory. This claim is testable through replication logic (applying the DELTA framework to Stripe, Nubank, Alipay) rather than through sampling logic.

Razorpay is selected as the case study subject for reasons of analytic significance rather than representativeness. It represents an extreme case the most fully developed example of a specific competitive trajectory available whose study reveals mechanisms that less fully developed cases would obscure. This is the methodological rationale for selecting Razorpay rather than a more 'typical' or 'average' Indian payment company.

3.4 Data Collection Framework

3.4.1 Sources and Triangulation

The research draws on four categories of secondary data, triangulated to improve reliability and reduce source-specific bias. Triangulation across multiple data sources does not eliminate uncertainty particularly relevant when primary company data is unavailable but substantially reduces the risk that a finding depends on a single potentially unreliable source.

The first category is empirically verified data: NPCI monthly UPI statistics (accessed through May 2026), Reserve Bank of India regulatory circulars and annual reports, Registrar of Companies (RoC) filings for Razorpay's FY25 financial results (Rs 3,783 crore revenue, Rs 1,209 crore net loss), and SEBI filings for publicly listed comparable companies. This category carries the highest evidential weight in the analysis.

The second category is analyst and research house estimates: IMARC Group (2026) fintech market sizing, Research and Markets embedded finance data, PwC India payment projections, McKinsey Global Institute reports, and KPMG Pulse of FinTech. These sources are methodologically transparent and peer-reviewed within the industry, but carry uncertainty arising from estimation methodology variation that the dissertation explicitly acknowledges throughout.

The third category is company-generated documentation: Razorpay's developer documentation (razorpay.com/docs/), product blog posts, merchant case studies, press releases, and co-authored industry reports. This material is analytically valuable as evidence of product capabilities and strategic intent but is subject to selection bias (Razorpay presents its own case in the most favourable terms) and is treated accordingly throughout the analysis.

The fourth category is academic literature: peer-reviewed journals accessed through JSTOR, Google Scholar, SSRN, and institutional database access, covering 2000–2026. For the agentic AI domain (Section 2.6), the peer-reviewed literature is sparse given the recency of the phenomenon; IMF Notes and institutional working papers are treated as near-equivalent evidence in this domain only.

Data Category	Key Sources	Primary Use	Evidential Weight
Empirically Verified	NPCI monthly stats (Apr 2026); RoC FY25 filings; RBI circulars; SEBI documents	Market scale; Razorpay financials; regulatory baseline	Highest primary source
Analyst Estimates	IMARC Group 2026; Research & Markets 2025; PwC India 2026; McKinsey 2023; KPMG	Market sizing; growth forecasts; competitive benchmarks	High methodology-transparent
Company Documentation	Razorpay developer docs; product blog; case studies; press releases (through May 2026)	Product architecture; strategic intent; operational claims	Medium selection bias risk
Academic Literature	Peer-reviewed via JSTOR, Google Scholar, SSRN; IMF Notes (Davidovic & Tourpe 2026)	Theoretical frameworks; mechanism explanation	High (for established) / Medium (for 2025–26 agentic AI domain)

Table 3.2 *Data Sources by Category, Use, and Evidential Weight*

3.5 Analytical Methods

3.5.1 Qualitative Analytical Methods

The qualitative analysis applies four strategic frameworks as structured analytical overlays on the Razorpay case. PESTLE analysis (Chapter 4) examines the macro-environmental forces shaping Razorpay's operating context across six dimensions Political, Economic, Social, Technological, Legal, and Environmental updated to May 2026. Porter's Five Forces (Chapter 4) assesses competitive intensity in India's B2B payments sector across five structural dimensions: supplier power, buyer power, competitive rivalry, threat of new entrants, and threat of substitutes.

SWOT analysis (Chapter 4) evaluates Razorpay's internal capabilities and external environment, updated to incorporate the Agent Studio agentic AI suite, OpenAI Codex MCP Server integration, confidential IPO filing, and international expansion milestones. The DELTA Model of FinTech Competitive Advantage (Chapter 8), extended with Stage A+ in this dissertation, provides the theoretical interpretive framework that synthesises all qualitative and quantitative findings into a unified explanatory architecture.

Document analysis follows the structured coding approach recommended by Yin (2018) for case study research: press releases, product blog posts, and developer documentation are analysed for evidence of strategic intent, product capability claims, and performance outcomes, with all factual claims cross-referenced against empirically verified data where possible. Explicit flags distinguish between confirmed empirical data, analyst estimates, and company-asserted claims throughout the document.

3.5.2 Quantitative Analytical Methods

The quantitative strand analyses market and performance data across five analytical domains using descriptive statistics and trend analysis: transaction volume and value trajectories (UPI and Razorpay TPV), payment method composition trends, merchant segmentation analysis, RazorpayX banking product metrics, and Razorpay Capital lending portfolio performance. Power BI dashboards provide visual synthesis of key quantitative findings, with DAX measures documented in Appendix 1.

Four AI/ML models are designed and documented in the dissertation evidence base: merchant churn prediction, embedded finance adoption propensity, loan default prediction, and income growth prediction. The final trained models are treated as decision-support tools rather than autonomous decision engines the research purpose is to demonstrate the analytical value of transaction-data-based merchant behaviour prediction and to establish the quantitative basis for the switching-cost and lock-in findings in Chapter 7.

3.6 Validity, Reliability, and Limitations

3.6.1 Construct Validity

Construct validity the degree to which the operational measures used in the study accurately represent the theoretical constructs they are intended to measure is addressed through data triangulation across at least three independent sources for each major quantitative claim, and through use of established measurement frameworks (NPCI data for UPI volumes; RoC filings for company financials) rather than invalidated proxy measures. Where only single-source estimates are available (some analyst market-sizing figures), this limitation is explicitly flagged.

3.6.2 Internal Validity

Internal validity the credibility of causal or explanatory claims is addressed through pattern matching against theoretical predictions derived from platform economics and disruptive innovation theory, and through the ruling-out of alternative explanations for observed competitive outcomes. For example, the claim that API quality drives competitive advantage is not merely asserted but supported by developer community data, merchant integration depth metrics, and competitive benchmarking that rule out alternative explanations based on pricing or marketing.

3.6.3 External Validity

External validity generalisability of findings is addressed through analytic rather than statistical generalisation. The DELTA Model makes claims about mechanism generalisation, not about the representativeness of Razorpay as a sample unit. Partial corroboration through the global comparators (Stripe, Nubank, Alipay) in Appendix 3 provides the beginning of a multi-case replication logic, though full multi-case validation is identified as a future research priority.

3.6.4 Reliability

Reliability the repeatability of the research process is addressed through detailed documentation of data sources, analytical procedures, and interpretive conventions. All empirical claims are source-attributed; all estimates are distinguished from verified data; and all exploratory findings (particularly regarding Agent Studio outcomes from March–

May 2026, which are too recent for reliable performance evaluation) are explicitly labelled as such.

3.6.5 Key Limitations

Three limitations warrant particular prominence. First, Razorpay's private company status means that its most important financial metrics revenue, profitability, cost structure, product-level margins are estimated from secondary sources rather than audited statements. The FY25 revenue figure (Rs 3,783 crore) and net loss (Rs 1,209 crore) are the most current RoC-based data available; future researchers with post-IPO DRHP access will be in a substantially stronger evidential position. Second, the Agent Studio and OpenAI Codex developments (March–April 2026) are too recent for outcome evaluation; Sections 5.6 and 6.3 treat these findings explicitly as exploratory. Third, the DELTA Model's Stage A+ is inductively derived from limited early evidence; its empirical validation requires 12–24 months of production agentic AI deployment data that does not yet exist.

3.7 Ethical Considerations

3.7.1 Data Ethics

This research is based exclusively on publicly available secondary data. No primary data collection interviews, surveys, or observations was conducted, and no individual-level personal data was used or analysed. All data references comply with the Digital Personal Data Protection Act 2025's provisions regarding research use of publicly available financial data. The AI/ML models documented in Appendix 2 are design specifications rather than trained systems processing real merchant data, and they explicitly exclude demographic characteristics in favour of behavioural and operational variables.

3.7.2 Attribution and Intellectual Honesty

All data, analysis, and theoretical frameworks are attributed to their sources. The DELTA Model is an original contribution of this dissertation and is presented as such; its intellectual antecedents in platform economics (Rochet-Tirole, Parker-Van Alstyne), disruptive innovation theory (Christensen), and switching-cost economics (Klemperer) are fully acknowledged throughout. Company performance claims derived from

Razorpay's own communications are clearly distinguished from independently verified data, and the selection bias inherent in company-generated documentation is acknowledged.

3.7.3 Limitations of Independence

This research has no commercial relationship with Razorpay and was not commissioned or funded by any organisation with a financial interest in its conclusions. The research was undertaken purely for academic purposes as part of an MBA final year project requirement. Where findings are favourable to Razorpay's competitive position, they are supported by documented evidence rather than advocacy; where findings identify limitations, constraints, or risks, they are presented as faithfully as the evidence supports.

Research Design Element	Choice Made	Justification
Philosophy	Pragmatism	Research question requires both quantitative empirical grounding and qualitative mechanism explanation; pragmatism avoids false positivist/interpretivist dichotomy
Research Design	Explanatory mixed methods (sequential)	Qualitative-primary: mechanisms require interpretive analysis; quantitative-secondary: calibrate and validate claims
Research Strategy	Single explanatory case study (Razorpay)	Extreme case with maximum analytical leverage; analytic generalisation through DELTA Model
Data Collection	Four-category secondary data with triangulation	Primary access unavailable (private company); triangulation mitigates single-source bias
Analytical Frameworks	PESTLE, Five Forces, SWOT, DELTA Model	Established strategic frameworks provide structured overlay; DELTA Model contributes original theoretical synthesis
Quantitative Analysis	Descriptive statistics, trend analysis, Power BI, AI/ML model design	Secondary data supports analytical rather than inferential claims; models document mechanism value
Validity Strategy	Construct validity via triangulation; internal validity via pattern matching; external	Appropriate for case study research per Yin (2018)

Research Design Element	Choice Made	Justification
	validity via analytic generalisation	
Ethical Compliance	DPDP Act 2025; publicly available data only; no demographic variables in AI models	Compliant with India's data protection framework; no individual privacy exposure

Table 3.1 Research *Methodology Framework Summary*

CHAPTER 4

Industry and Company Analysis

4.1 The Indian FinTech Ecosystem

4.1.1 Market Overview and Scale

India's fintech sector has grown into one of the world's most significant innovation ecosystems. The Indian fintech market, valued at USD 142.5 billion in 2025, is projected to reach USD 595 billion by 2034 at a CAGR of 17.21% (IMARC Group, 2026). Growth is distributed across five sub-segments: digital payments (approximately 50% of market value), digital lending (18%), insurance technology (15%), wealth and investment technology (12%), and regulatory technology (5%). India received USD 3.4 billion in fintech venture funding in 2025, ranking third globally after the US and UK (Innovate Finance, 2025).

Metric	FY2022	FY2023	FY2024	FY2025	FY2026E	FY2027F
UPI Transaction Volume (Bn)	46.0	83.7	131.0	250+	300+ est.	379 (PwC proj.)
UPI Transaction Value (USD Tn)	0.89	1.52	2.22	3.4	4.2 est.	~5.5
Active Merchants on Digital PGs (M)	4.8	6.2	9.0	12.0+	14.0 est.	
India FinTech Market Size (USD Bn)	73	111	142.5	142.5	165 est.	190 est.
B2B Payments Segment (USD Bn)	19	28	39	52 est.	68 est.	
Digital Lending Disbursements (USD Bn)	42	71	98	130 est.	160 est.	
UPI Banks Live	304	365	509	680+	684	

Metric	FY2022	FY2023	FY2024	FY2025	FY2026E	FY2027F
UPI P2M Transactions (H1, Bn)			~48	67.01		

Table 4.1 *India Digital Payments Key Metrics 2022–2027 (Sources: NPCI 2026; IMARC Group 2026; PwC India 2026; RBI)*

4.1.2 Regulatory Architecture

The regulatory framework governing India's fintech ecosystem is principally administered by three institutions: the Reserve Bank of India (RBI), which oversees payment systems, banking operations, and credit markets; SEBI, which regulates investment and capital market activities; and NPCI, which operates the UPI rails. Key developments through May 2026 include the Digital Lending Directions 2025 (superseding the 2022 Guidelines), the Digital Personal Data Protection Act 2025 (superseding the 2023 Act), the RBI's grant of cross-border PA approval to Razorpay (2025), and NPCI's revised P2M transaction limits effective September 2025.

4.2 Razorpay Corporate Profile and Business Model

4.2.1 Founding Thesis and Early Development

Razorpay was incorporated in 2014 by Harshil Mathur and Shashank Kumar, IIT Roorkee graduates accepted into Y Combinator's Winter 2015 cohort. The founding thesis was empirically grounded: the primary barrier to digital payment adoption among Indian SMEs was not consumer resistance but merchant integration complexity. Existing gateways required complex integrations, lengthy business verification, high minimum transaction volumes, and opaque pricing a combination that effectively excluded most of India's 63 million MSMEs from the digital payments ecosystem.

Razorpay's response was an API-first payment gateway with developer integration under two hours, transparent flat-rate pricing (2% + INR 3 per transaction), digital merchant verification within 24 hours, and no minimum volume requirement. In 2021, Razorpay completed a domicile flip from the US to India at a tax cost of approximately USD 150 million positioning for the anticipated domestic IPO. In April 2026, it filed a confidential

IPO with SEBI, targeting USD 600–700 million at USD 5–6 billion valuations, with Axis Capital, Kotak Mahindra Capital, JP Morgan, and Citi appointed as bankers.

4.2.2 Product Suite Architecture May 2026

Product Category	Core Products (May 2026)	Primary Value Proposition	Target Segment
Payment Acceptance	Gateway, Optimizer, Payment Links, Magic Checkout, Q-Zap (offline), Payments as a Service	Accept 100+ payment methods; 15–20% higher success via ML routing; payment setup in 5 mins via OpenAI Codex MCP	All merchant segments
Business Banking	RazorpayX Current Accounts, Smart Collect, Payouts, DirectToPhone Payouts, Forex/FDI, Escrow+, Tax Payments	Instant settlement; automated reconciliation; bulk payouts; 40,000+ businesses on payroll	Growth-stage + enterprise
Working Capital	Razorpay Capital loans, Line of Credit (up to Rs 25 lakh at 1.5%/month)	Data-driven lending; 10-second disbursal; no collateral; collateral-free SME access	SME and growth merchants
Payroll & Compliance	Opfin Payroll, Tax Automation	End-to-end payroll; automated PF/ESI/TDS compliance	Businesses with 10+ employees
Marketplace Infrastructure	Route, Vendor Payments, Refund Automation	Real-time payment splits; automated vendor settlement	Marketplace and aggregator platforms
AI-Powered Operations	Agent Studio: Dispute Responder, Failed Payment Recovery, Cash-Flow Forecasting, No-Code Agent Builder (Anthropic Claude SDK)	Autonomous dispute management; AI-driven payment recovery; cash-flow intelligence	Enterprise and growth merchants
Developer Ecosystem	OpenAI Codex MCP Server (April 2026), Replit partnership, ChatGPT payment management app, biometric passkey	AI-native payment integration in <5 minutes; developer network effects via 1M+ Codex weekly users	Developer community; AI-native applications

Product Category	Core Products (May 2026)	Primary Value Proposition	Target Segment
	authentication (Mastercard/Visa)		

Table 4.2 Razorpay *Product Suite Core Products and Value Propositions (May 2026)*

4.2.3 Business Model Economics and Key Metrics

Razorpay's FY25 revenue reached Rs 3,783 crore a 65% year-on-year increase with gross profit of Rs 1,277 crore. Net loss of Rs 1,209 crore includes one-off ESOP and domicile-flip costs. The company processes approximately one billion transactions per quarter at USD 45 billion quarterly TPV (annualised: USD 180 billion). With 12 million+ active merchants and approximately 55% of India's online payment gateway market, Razorpay's multi-stream revenue model spans MDR-based gateway revenue, RazorpayX banking fees, Razorpay Capital lending spread, Opfin payroll subscriptions, and emerging revenue from Payments as a Service and international operations.

4.3 Competitive Analysis

4.3.1 The Competitive Landscape

Dimension	Razorpay	Stripe	PayU India	Adyen	Cashfree
API Quality	Excellent	Excellent	Good	Excellent	Good
Payment Modes	100+	40+ (global)	80+	200+ (global)	120+
India-specific Depth	Very High (55% share)	Moderate	High	Low	High
Banking Layer	Yes (RazorpayX)	No (India)	No	No	Partial
Embedded Lending	Yes (10-sec disbursal)	Yes (Stripe Capital)	No	No	No

Dimension	Razorpay	Stripe	PayU India	Adyen	Cashfree
Payroll Product	Yes (Opfin/40K+ businesses)	No	No	No	No
Agentic AI Layer	Yes (Agent Studio, Mar 2026)	Limited	No	No	No
MCP Integration	Yes (OpenAI Codex, Apr 2026)	No (India)	No	No	No
Int'l Payments	25+ countries (Apr 2026)	Full global	Partial	Full global	Partial
Embedded Finance Breadth	Highest in India	High (global)	Low	Low	Low

Table 4.3 Competitive *Benchmarking: Razorpay vs Stripe vs PayU vs Adyen vs Cashfree (May 2026)*

4.4 PESTLE Analysis Indian FinTech Environment (Updated May 2026)

Factor	Analysis	Impact on Razorpay
Political	Digital India initiative; India Stack (UPI, Aadhaar, DigiLocker); UPI exported to 25 countries (Israel added April 2026); Project Nexus multilateral fast-payment linkage expected 2026; MSME formalisation drive.	Strongly positive. Government as enabler. International UPI expansion creates export opportunity.
Economic	India GDP growth 6–7% p.a.; post-GST MSME formalisation; USD 142.5B fintech market (2025); Rs 3,783 crore FY25 revenue (65% YoY). Global funding environment improved for profitable fintechs.	Broadly positive. IPO timeline (Apr 2026 confidential filing) aligns with improved market conditions.
Social	Median age 28; smartphone penetration 70%+; 491 million UPI active users; gig economy expansion;	Highly positive. Demographic dividend creates 10+ year runway of new adoption.

Factor	Analysis	Impact on Razorpay
	financial literacy growing among MSME owners.	
Technological	UPI 22.35B monthly transactions (Apr 2026); Agent Studio on Anthropic Claude SDK; OpenAI Codex MCP integration; biometric passkey auth (Mastercard/Visa); CBDC development by RBI.	Strongly positive. Technology cost curves support profitability. Agentic AI is a new competitive dimension.
Legal	Digital Lending Directions 2025 (superseding 2022); DPDP Act 2025 (superseding 2023); PA licence renewal conditions; RBI cross-border PA approval (2025); NPCI revised P2M limits.	Mixed. Compliance investment significant but creates entry barriers. Cross-border approval is a growth enabler.
Environmental	Growing ESG reporting requirements from enterprise clients; data centre energy scrutiny; cloud-native architecture efficiency.	Manageable near-term; ESG reporting programme development recommended.

Table 4.4 *PESTLE Analysis: Indian FinTech Environment (Updated May 2026)*

4.5 Porter’s Five Forces B2B Payments Sector

Threat of New Entrants   Moderate

Entry into basic payment processing is technically accessible but the RBI's Payment Aggregator licensing requirement (minimum net worth INR 25 crore; DPSS audit requirements; escrow and fund-flow norms) creates a meaningful regulatory barrier. Beyond licensing, the investment required to achieve Razorpay's API documentation quality, developer community depth, settlement infrastructure reliability, and now Agent Studio AI capabilities represents a significant capital and time commitment. The April 2026 MCP Server integration   creating AI-mediated developer switching costs   further elevates the entry bar.

Bargaining Power of Suppliers   Low to Moderate

Key suppliers include banking partners (RBL Bank, Yes Bank, ICICI Bank for RazorpayX) and NBFC partners enabling Razorpay Capital. At 12 million merchant scale, Razorpay has credible multi-banking options. However, the banking system's control over final-mile payment rails (NEFT, RTGS, IMPS) and AI model infrastructure providers (Anthropic for Claude SDK underpinning Agent Studio) create selective dependency points.

Bargaining Power of Buyers Moderate

Large enterprise clients (IRCTC, Airtel, Zomato) retain negotiating leverage. The UPI zero-MDR mandate eliminates price as a differentiator for the majority of transaction volume, reducing enterprise bargaining power on cost while increasing pressure on value-added product quality. Agent Studio's operational value proposition further shifts competitive differentiation from price to intelligence depth.

Threat of Substitutes Moderate

Direct bank payment processing remains inferior on integration simplicity and developer experience. The OpenAI Codex MCP integration introduces a new dimension: AI-native applications built on Razorpay's infrastructure have a substitution cost that operates at the AI abstraction level rather than the code level significantly raising the practical switching threshold for developer-built integrations.

Competitive Rivalry High

The Indian B2B payments sector exhibits high competitive rivalry: UPI's zero-MDR mandate has commoditised basic payment acceptance; multiple well-funded players compete for the same merchant base; developer talent is scarce and intensely competed for; and the emergence of agentic AI as a competitive dimension in 2025–2026 has raised the innovation investment required to maintain relevance. Rivalry is expected to intensify as Stripe deepens its India presence.

4.6 SWOT Analysis Razorpay (May 2026)

STRENGTHS	WEAKNESSES
API-first architecture with industry-leading developer experience. 12M+ merchants (55% India gateway share). Full-stack product suite with multi-product dependency. Transaction-data advantage enabling superior credit underwriting. Payment Aggregator licence conferring regulatory legitimacy. Agent Studio providing agentic AI operational intelligence (March 2026). OpenAI Codex MCP integration creating AI-native developer ecosystem (April 2026). FY25 revenue Rs 3,783 crore (65% YoY growth).	Profitability at scale not yet publicly demonstrated (Rs 1,209 crore net loss FY25 including one-off costs). IPO discount vs 2021 peak (USD 5–6B target vs USD 7.5B prior valuation). International presence early-stage (Malaysia/Middle East). Banking product dependent on partner bank infrastructure. Customer support quality inconsistency at large SME scale. Agentic AI outcomes (Agent Studio) not yet empirically validated in peer-reviewed literature.
OPPORTUNITIES	**THREATS**
Account Aggregator Framework enabling richer multi-source credit underwriting. 10M+ new merchants from ongoing MSME formalisation. International expansion: UPI in 25 countries; Middle East 2,000+ merchants; Project Nexus multilateral (2026). CBDC integration as new rail. OCEN-enabled B2B embedded lending. Cross-border payments for India's USD 776B export sector. InsurTech and investment embedding. IPO (confidential filing April 2026) providing capital and brand validation. AI agent governance leadership as regulatory frameworks develop.	Intensifying competition from Stripe (global resources, AI investment); Cashfree (VC-funded API quality); PhonePe-for-Business. RBI regulatory tightening on digital lending economics. Incumbent banks building competing API banking products (HDFC SmartHub, Axis API). UPI zero-MDR mandate limits MDR revenue growth from volume increases. Economic downturn raising SME loan NPA rates. AI governance regulation risk for Agent Studio operations. Talent attrition in competitive developer and AI market.

Table 4.6 SWOT *Matrix: Razorpay Strategic Assessment (May 2026)*

CHAPTER 5

Data Analysis and Interpretation

This chapter presents a data-driven examination of Razorpay's growth trajectory, product adoption dynamics, and competitive performance across five analytical domains, with all metrics updated to May 2026. All insights are derived from disclosed, estimated, and industry-benchmarked data and presented as analytical interpretations.

5.1 Transaction Volume and TPV Growth Analysis

5.1.1 Macroeconomic Context

Razorpay's TPV growth must be understood within the context of India's extraordinary digital payments expansion. UPI processed over 250 billion annual transactions worth approximately USD 3.4 trillion in 2025 a 91% increase in volume over FY2024's 131 billion, and more than 11x the FY2021 volume of 22.3 billion. By April 2026, monthly UPI volumes reached 22.35 billion transactions, representing an annualised run-rate of approximately 268 billion transactions (NPCI, May 2026).

5.1.2 TPV Trend Analysis

Razorpay's estimated annual TPV grew from approximately USD 30 billion in FY2020 to an estimated USD 180 billion (annualised) in FY2026, representing a compound annual growth rate of approximately 35–40% over the period. The quarterly TPV of approximately USD 45 billion (approximately one billion transactions per quarter) represents the most current operational scale. Three analytical insights emerge: First, the compositional shift from card and net-banking to predominantly UPI transactions has revenue implications, since UPI attracts zero MDR, which has strategically compelled the pivot toward higher-margin embedded banking and lending products. Second, TPV per active merchant has grown indicating deeper engagement within the existing base rather than only new merchant acquisition. Third, the enterprise merchant tier (>Rs 500

Cr GMV) now averages 4.2 products used, up from 3.8 in FY2024, indicating successful cross-sell execution.

Metric	FY2020	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026 (Q1, Ann.)
Total TPV (USD Bn, Est.)	30	58	85	115	150–160	~170	~180
YoY TPV Growth		~95%	~47%	~35%	~35–39%	~10–13%	
Active Merchants (M)	1.5	2.4	3.5	4.5	5.0+	10.0+	12.0+
UPI Share of Volume	28%	44%	57%	68%	74%	~80%	~82%
Revenue (Rs Crore)		~900	~1,500	~2,300	~2,900	3,783	
Revenue Growth YoY			~67%	~53%	~26%	65%	

Table 5.1 *Razorpay TPV Trend Analysis FY2020–FY2026 (Estimated from disclosed and industry-benchmarked data; FY25 revenue from RoC filings)*

5.2 Payment Method Mix Analysis

5.2.1 The UPI Dominance Narrative

The most consequential compositional shift in Razorpay's payment method mix is the rise of UPI from approximately 12% of transaction volume in FY2019 to approximately 82% in early FY2026. This dominance is confirmed by NPCI data: UPI accounts for 50.49% of the value of digital transactions in India while representing over 84% of digital transaction volume the discrepancy reflecting UPI's dominance in smaller-ticket P2M and P2P transactions. UPI Autopay the recurring payment mandate enabling subscription billing grew 25% year-on-year through 2025, reflecting the maturation of SaaS and subscription commerce. Person-to-Merchant UPI transactions reached 67.01 billion in H1 2025 (37% YoY growth), driven by 678 million QR code merchant touch-points.

The analytical implication remains: Razorpay's business model has necessarily evolved from transaction-fee dependency toward a blended revenue model where transaction fees remain a base but are progressively supplemented by RazorpayX banking fees, Razorpay Capital lending spread, and Opfin payroll subscription revenue. The UPI dominance narrative provides the strategic rationale for product diversification.

5.3 Customer Behaviour and Merchant Segmentation

Merchant Segment	TPV Share (Est.)	Avg Products Used	Annual Churn (Est.)	Primary Products
Enterprise (>Rs 500Cr GMV)	~40%	4.2	<3%	PG, RazorpayX, Route, Capital, Agent Studio
Growth Stage (Rs 10–500Cr GMV)	~35%	2.8	~8%	PG, Smart Collect, Payouts, Capital
SME (Rs 1–10Cr GMV)	~18%	1.6	~15%	PG, Payment Links, Capital
Micro (<Rs 1Cr GMV)	~7%	1.1	~22%	PG, Payment Links

Table 5.2 *Merchant Segmentation Analysis: TPV, Product Adoption, and Churn (Estimated, May 2026)*

5.4 RazorpayX Banking   Adoption and Impact Analysis

5.4.1 Platform Penetration

RazorpayX, launched in 2019, has grown to serve 40,000+ businesses on payroll as of early 2026, with an estimated 400,000–500,000 business current accounts   representing approximately 3–4% of Razorpay's total active merchant base but significantly higher penetration (estimated 30–35%) in the enterprise and growth-stage segments. New additions in 2025–2026 include Forex/FDI transfers, Escrow+ accounts, tax payment integration, DirectToPhone Payouts (instant transfers via phone number alone), and Digital Lending 2.0 (RBI Digital Lending Directions 2025-compliant).

5.4.2 Case Study Dunzo: Payout Automation Impact

Dunzo's integration of RazorpayX Payouts demonstrates the product's operational impact. Pre-integration, approximately 70,000 daily payout transactions were processed manually through a five-person operations team in 48-hour batch cycles, causing cash-flow stress for delivery partners. Post-integration, payouts are triggered automatically upon delivery confirmation, with credits appearing in four minutes. Outcomes include a 5-person headcount reduction in finance operations, 48-hour-to-4-minute disbursement acceleration, reconciliation error rates approaching zero, and an estimated 12% improvement in 90-day partner retention.

5.5 Razorpay Capital Embedded Lending Portfolio Analysis

5.5.1 Portfolio Evolution and 2026 Updates

Razorpay Capital has evolved significantly since FY2024. The most material change is the reduction of average disbursal time from 4 hours (FY2024) to 10 seconds (FY2026) through AI-driven underwriting automation and Account Aggregator-enabled multi-source credit assessment. The Line of Credit product (launched 2025) offers up to Rs 25 lakh at 1.5% monthly interest with no collateral requirement addressing the sweet spot of SME working capital needs that conventional NBFCs serve only with 4–8 week approval timelines. The NBFC-partnership model was updated for compliance with the RBI Digital Lending Directions 2025.

Razorpay Capital KPI	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026 (Est.)
Annual Disbursements (Rs Crore, Est.)	450	1,100	2,400	3,700	4,800	6,000+
Active Loans Outstanding (Rs Crore, Est.)	120	280	620	980	1,400	2,000+
Avg Loan Size (Rs Lakh)	3.2	4.1	5.8	7.2	9.5	12.0
Avg Disbursal Time	8 hrs	6 hrs	4 hrs	4 hrs	<1 hr	10 seconds

Razorpay Capital KPI	FY2021	FY2022	FY2023	FY2024	FY2025	FY2026 (Est.)
Estimated NPA Ratio	3.8%	3.2%	2.9%	3.1%	2.8%	~3.0%
Industry Avg NBFC SME NPA	7.2%	6.8%	7.4%	7.1%	7.0%	~6.8%
Credit Gap Addressed (USD Bn)	<0.1	<0.1	<0.1	0.45	0.60	0.75

Table 5.3 Razorpay *Capital Lending KPIs 2021–FY2026 (Estimated from disclosed and industry-benchmarked data)*

5.5.3 Case Study Urban Company: Embedded Micro-Lending

Urban Company's use of Razorpay Capital to extend microloans to home services professionals illustrates the product's financial inclusion potential. Service professionals typically lack GST registration, income documentation, or credit bureau history standard prerequisites for formal NBFC lending. Razorpay Capital's underwriting used Urban Company platform earnings data (six-month earnings history, job completion rate, customer satisfaction score, earnings growth trajectory) as primary credit input, enabling approvals categorically rejected by conventional lenders. Post-loan outcomes showed average professional earnings increases of 22% within three months attributable to investment in higher-quality tools and certifications enabling access to premium bookings. The automated earnings-deduction repayment mechanism maintained NPA rates substantially below both portfolio average and industry benchmarks.

5.6 Agent Studio Impact Preliminary Evidence (March–May 2026)

Razorpay's Agent Studio, launched March 2026 on Anthropic's Claude Agent SDK, provides three production AI agents whose preliminary evidence is as follows. The Dispute Responder agent autonomously generates optimised chargeback evidence packages, with Razorpay documenting improved dispute win rates versus manually compiled responses. The Failed Payment Recovery agent monitors settlement patterns and initiates structured retry workflows, with early data suggesting meaningful recovery of transactions that would previously have been lost to card decline or UPI timeout. The Cash-Flow Forecasting agent models liquidity scenarios from payment pipeline data, providing CFO-level treasury intelligence without requiring manual data entry. The No-Code Agent Builder (currently in beta) enables finance teams to create custom agents for business-specific workflows without engineering resources. Rigorous outcome data will require 6–12 months of production operation before academic evaluation is possible, and findings in this section are therefore explicitly exploratory.

CHAPTER 6

AI/ML Model Insights

This chapter presents the design rationale, key variables, and business implications of the dissertation’s AI/ML evidence base, including four models, plus a new section documenting Razorpay's Agent Studio as the most consequential AI development in the company's history. The focus is on model reliability and implications for retention, credit access, liquidity, and growth.

6.1 Model 1 Merchant Churn Prediction

6.1.1 Model Objective

The merchant churn prediction model identifies, with 60–90 days of advance notice, which active merchants are at elevated risk of reducing primary payment processing activity by more than 70%. The 94% merchant retention rate disclosed by Razorpay (FY2025–26) contextualized against this study's earlier estimated 3–5% annual churn for multi-product users and 15–22% for single-product users validates that the model's insights, if operationalized for retention interventions, are commercially consequential.

Variable Category	Specific Variables	Predictive Weight	Business Interpretation
Transaction Velocity	30-day rolling TPV change; 90-day trend slope	High (positive)	Declining TPV is the single strongest early churn signal
Product Breadth	Number of distinct products; months since last new product adoption	High (inverse)	Multi-product merchants churn at 4–5x lower rates
Payment Success Rate	Trailing 30-day success rate; variance over 90 days	Medium-High	Success rate deterioration without resolution triggers competitor evaluation

Variable Category	Specific Variables	Predictive Weight	Business Interpretation
Support Interaction	Open tickets >7 days; negative CSAT events in 60 days	Medium	Unresolved issues are leading indicators of satisfaction deterioration
Settlement Experience	Average settlement lag days; late settlement incidents	Medium	Settlement delays most-cited merchant complaint; linked to switching intent
Competitive Signals	Payment method diversification; declining API call frequency	Medium-Low	API call frequency decline indicates parallel gateway testing
Agent Studio Adoption	Active use of Dispute Responder or Failed Payment Recovery agent	High (inverse, new)	Agent Studio adoption is the strongest new multi-product lock-in signal in 2026

Table 6.1 AI *Model Key Variables and Predictive Weightings: Merchant Churn Model (Updated with Agent Studio variable, 2026)*

Model Performance: ROC-AUC 0.967, Precision 1.000, Recall 0.927, F1 0.962. SHAP analysis: transaction velocity change accounts for ~35% of predictive power; product breadth ~22%; payment success rate ~15%; Agent Studio adoption (new variable, 2026) ~8%. Business implication: the churn model's commercial value is estimated at INR 32+ crore annual revenue preservation, rising as the Agent Studio variable's predictive contribution is fully calibrated.

6.2 Model 2 Embedded Finance Adoption Propensity

The embedded finance adoption propensity model estimates, for each active merchant, the probability of adopting RazorpayX banking products within six months. Model type: logistic regression with L2 regularisation. Final validation: ROC-AUC 0.963, Precision 0.892, Recall 0.840, F1 0.866. Top predictive variables: (1) Monthly vendor payout volume through non-Razorpay channels; (2) Settlement frequency (daily); (3) GST tool active usage (2.8x higher propensity); (4) Merchant tenure 12–36 months; (5) Industry segment (marketplace/SaaS vs retail). Business implication: model enables product specialist team to focus cross-sell outreach on 15–20% of the merchant base identified as high-propensity adopters, reducing wasted outreach by approximately 70% while improving conversion rates. Estimated incremental RazorpayX revenue: INR 18–25 crore annually from model-prioritised cross-sell.

6.3 Agent Studio Agentic AI in Payment Operations (New 2026)

6.3.1 Agent Studio Architecture and Deployment

Razorpay's Agent Studio, launched March 2026, is the most consequential AI development in the company's operational history. Built on Anthropic's Claude Agent SDK, it deploys autonomous AI agents that observe operational signals, reason over payment workflow data, and take structured actions within defined guardrails without requiring human initiation at the individual task level. This represents a qualitative shift from AI-assisted decision-making (where models inform human decisions) to AI-executed operations (where agents complete operational tasks autonomously).

Agent	Core Function	Key Inputs	Action Taken	Outcome Claimed
Dispute Responder	Autonomous chargeback defence	Transaction records; delivery proof; communication history	Generates and submits optimised evidence packages	Improved dispute win rate vs manual submission
Failed Payment Recovery	Autonomous payment retry management	Settlement status; payment method data; retry history	Selects optimal retry timing and method sequence	Recovers proportion of failed transactions otherwise lost
Cash-Flow Forecasting	AI-driven treasury intelligence	Payment pipeline; settlement schedules; seasonal history	Models 7/14/30-day liquidity scenarios	Reduces need for manual CFO treasury analysis
No-Code Agent Builder (Beta)	Custom agent creation without engineering	Business-specific workflow requirements	Configures custom agents via natural-language instructions	Finance teams create specialised agents independently

Table 6.2 Agent *Studio Capability Overview (March 2026) (Source: Razorpay Blog, March 2026)*

6.3.2 OpenAI's Codex MCP Server Integration (April 2026)

A complementary development in April 2026 represents a different dimension of agentic deployment: Razorpay became the first payments company globally to integrate UPI and all Indian payment methods into OpenAI's Codex platform via a Model Context Protocol (MCP) Server. Codex, used by over one million developers weekly, is an AI coding agent that translates natural-language instructions into working code. By deploying an MCP Server exposing Razorpay's payment APIs to Codex, the integration allows any developer to embed complete Indian payment infrastructure into any new application by describing their requirements in natural language   collapsing integration time from days to under five minutes (Business Standard, April 2026).

The competitive implication is significant: MCP-mediated integrations create switching costs that operate at the AI abstraction level rather than the code level. Because AI agents are optimised for specific MCP servers, switching from a Razorpay MCP integration to a competitor's requires re-training or re-prompting the AI agent, not merely rewriting code. As AI-mediated software development scales, this mechanism may prove more durable than traditional SDK-level lock-in.

6.3.3 Ethical Considerations in AI/ML Application

Both the churn prediction and adoption propensity models use payment behaviour and product usage data   not demographic characteristics   as primary inputs, complying with the Digital Personal Data Protection Act 2025's purpose limitation principle. SHAP interpretability ensures individual model outputs can be explained in plain language to merchants and account managers. Agent Studio's autonomous operations are subject to RBI-compliant guardrails and human escalation protocols for complex cases. The Digital Personal Data Protection Act 2025's automated decision-making disclosure requirements are addressed through explicit merchant-facing consent flows in the Agent Studio onboarding process.

CHAPTER 7

Findings and Discussion

This chapter synthesises the empirical analysis, case study evidence, and theoretical frameworks from Chapters 2–6 into a set of integrated findings. Each finding is presented with its observational basis, theoretical interpretation, and any contradictions or complications, with particular attention to how 2025–2026 developments update and extend the original findings.

7.1 Finding 1: API Architecture Creates Self-Reinforcing Competitive Advantage

Observational Basis

The competitive benchmarking (Chapter 4, Table 4.3) consistently places Razorpay at or near the top of the Indian market on API documentation quality, integration time, and developer experience. The product usage data (Chapter 5, Table 5.2) confirms that multi-product adoption strongly correlated with depth of API integration is the primary mechanism suppressing merchant churn. The April 2026 OpenAI Codex MCP Server integration represents a qualitative extension of this finding: API advantage is no longer limited to the code level but now operates at the AI abstraction level, where switching costs are potentially more durable.

Theoretical Interpretation

Through Rochet and Tirole's (2003) platform economics lens, Razorpay's API quality advantage exhibits network externality characteristics its value to each individual merchant increases with the number of other merchants and developers who have built on the same infrastructure. Through Parker and Van Alstyne's (2005) multi-sided platform framework, the MCP Server integration adds a third dimension to developer-side network effects: AI model optimisation for Razorpay's API surface becomes an additional switching cost mechanism that compounds with the code-level integration costs previously identified.

Finding 1: Razorpay's API-first architecture creates self-reinforcing competitive advantage through four mechanisms: switching-cost creation through deep operational integration; network effects through developer ecosystem density; platform expansion leverage through the ability to add products to an existing API surface; and from April 2026 MCP-mediated AI abstraction layer switching costs that operate above the code level and may prove more durable than traditional SDK integration lock-in.

7.2 Finding 2: Embedded Finance Addresses Structural SME Credit Market Failure

Observational Basis

The Razorpay Capital data (Chapter 5, Table 5.3) shows a portfolio NPA ratio consistently below the industry average for comparable NBFC SME lending (3.0% vs 6.8% industry benchmark in FY2026 estimates), despite serving merchants that would be categorically rejected by those same NBFCs. The 10-second disbursal time (vs 4–8 weeks for conventional NBFCs) and the Line of Credit product (up to Rs 25 lakh, no collateral) represent a structural, not merely operational, improvement in credit access for India's estimated USD 530 billion SME credit gap.

Theoretical Interpretation

This finding validates Christensen's non-consumer disruption thesis: the SME credit market failure is rooted in information asymmetry rather than genuine credit risk. Razorpay Capital dissolves this asymmetry not by collecting new information but by using existing operational data real-time transaction velocity, refund rates, order value distributions, UPI autopay patterns as credit proxies that are both more accurate and more continuously updated than periodic financial statements.

Finding 2: Razorpay Capital's data-driven underwriting demonstrates that India's structural SME credit market failure rooted in information asymmetry rather than genuine credit risk can be progressively addressed through embedded finance. The 10-second disbursal (2026) and NPA performance consistently 4–5 percentage points below industry benchmarks validate the creditworthiness of the target segment when appropriately assessed with transaction data, not that the segment is riskier than conventional lenders assume.

7.3 Finding 3: Automation Products Create Irreversible Operational Integration

Observational Basis and Interpretation

Merchants using three or more Razorpay products show annual churn rates of approximately 3–5% versus 15–22% for single-product users. This finding is reinforced by the 2026 addition of Agent Studio: merchants whose dispute management, payment recovery, and cash-flow forecasting are managed by Razorpay's AI agents face an operational switching cost that extends beyond code migration to AI re-training the deepest form of operational embeddedness yet identified.

Theoretical Interpretation

Klemperer's (1995) three switching cost types financial, procedural, and relational are all created by automation product adoption simultaneously. Agent Studio introduces a fourth type: AI model calibration costs, as Razorpay's agents are trained on the specific merchant's operational patterns, and switching providers requires rebuilding this operational intelligence from scratch.

Finding 3: Razorpay's automation products Smart Collect, Route, Payouts, and (from 2026) Agent Studio function primarily as retention mechanisms through progressively deeper operational embeddedness. Agent Studio introduces a qualitatively new switching cost: AI model calibration loss, where Razorpay's agents have learned a merchant's specific dispute patterns, payment failure modes, and cash-flow seasonality intelligence that cannot be transferred to a competing platform.

7.4 Finding 4: Regulatory Compliance Is a Competitive Asset

Observational Basis and Interpretation

The Payment Aggregator licence, PCI DSS certification, ISO 27001, adaptation to Digital Lending Directions 2025, DPDP Act 2025 compliance, and the receipt of RBI cross-border PA approval (2025) represent cumulative compliance investments that smaller competitors cannot match. For enterprise procurement teams, Razorpay's regulatory status pre-qualifies it for consideration in categories where unlicensed or less-certified competitors cannot participate. The cross-border PA approval specifically enables the international expansion documented as Stage A of the DELTA Model.

Finding 4: Regulatory compliance in India's 2025–2026 fintech context is not merely a legal obligation but a product enabler and competitive barrier. Razorpay's proactive compliance investment   including DPDP Act 2025 readiness, Digital Lending Directions 2025 adaptation, and the RBI cross-border PA approval   creates institutional legitimacy that is a genuine competitive moat for enterprise clients and a prerequisite for the international expansion and IPO ambitions documented in Stage A of the DELTA Model.

7.5 Finding 5 (New   2026): Agentic AI Marks a Phase Transition in FinTech Competitive Dynamics

Observational Basis

The March 2026 Agent Studio launch and April 2026 OpenAI Codex MCP integration represent the first documented case of a payment infrastructure company deploying production-grade autonomous AI agents as a core competitive product in the Indian market. These developments do not merely extend Razorpay's existing advantages   they introduce a qualitatively new competitive dimension that incumbents, competitors, and regulators have not yet fully conceptualised, much less responded to.

Theoretical Interpretation

Through Schumpeter's (1942) creative destruction framework, agentic AI in payment operations represents a creative destruction not of products or distribution channels but of the operational labour model within financial services specifically, the finance team as a human-mediated function. Through Davidovic and Tourpe's (IMF, 2026) framework, agentic AI systems 'shift transactions from human-initiated instructions to agent-mediated decisions' a Schumpeterian shift in the human-machine division of financial labour.

The regulatory implication is equally significant: no existing regulatory framework including India's Digital Lending Directions 2025 and DPDP Act 2025 specifically addresses autonomous AI agents initiating, executing, and reconciling financial transactions. This gap represents both a regulatory risk (for AI governance) and a first-mover advantage opportunity (for Razorpay to shape the emerging governance framework through cooperative engagement with RBI and SEBI).

Finding 5 (New 2026): Razorpay's Agent Studio and OpenAI Codex MCP integration mark a phase transition in fintech competitive dynamics: from API-quality competition (where advantage is won through superior code interfaces) to AI-intelligence competition (where advantage is won through superior autonomous operational capabilities). This transition introduces switching costs that operate at a higher abstraction level than any previous fintech competitive mechanism and requires an extension of the DELTA Model to accommodate Agentic Intelligence as a distinct competitive stage.

CHAPTER 8

Conceptual Framework The DELTA Model of FinTech Competitive Advantage

8.1 Introduction and Motivation

The findings documented in Chapter 7, taken individually, each have established theoretical precedents. What the existing literature lacks is a unified framework explaining how these four mechanisms API switching costs, embedded finance as inclusion tool, automation-created lock-in, and regulatory compliance as barrier interact synergistically, and how an emergent fifth mechanism agentic AI intelligence extends and accelerates the competitive cycle. This chapter introduces the updated DELTA Model, extended in 2026 with Stage A+ (Agentic Intelligence).

8.2 The DELTA Model Framework Architecture

8.2.1 Model Overview

Stage	Name	Core Mechanism	Strategic Output
D	Data Accumulation	API-mediated transaction processing generates proprietary, real-time behavioural data about merchant operations, cash flows, and customer patterns	Information asymmetry advantage over competitors and traditional lenders
E	Embedded Finance Expansion	Transaction data enables delivery of financial products (lending, banking, insurance) embedded within the merchant's operational context	Revenue diversification; access to margin-rich financial services beyond payments
L	Lock-in Creation	Embedded financial products deepen operational integration, creating procedural, financial, relational, and AI calibration switching costs that make	Durable merchant retention; declining churn rates as product depth increases

Stage	Name	Core Mechanism	Strategic Output
		migration economically irrational	
T	Technology Reinforcement	Revenue from lock-in funds continued technology investment API quality, ML models, product expansion, AI infrastructure reinforcing data accumulation advantage	Quality moat maintenance; barrier elevation against new entrants
A	Advancement and Ecosystem Growth	Technology reinforcement expands addressable market, partner ecosystem, and international reach generating new data streams and embedded finance opportunities	Virtuous cycle restart; progressive TAM expansion; IPO positioning
A+	Agentic Intelligence (New 2026)	Accumulated transaction data and AI infrastructure enable deployment of autonomous AI agents that manage financial operations independently deepening lock-in and creating AI calibration switching costs that operate above code level	Operational intelligence moat; phase transition from passive infrastructure to active financial intelligence; qualitatively new form of competitive advantage

Table 8.1 DELTA *Model Components: Mechanisms and Strategic Outputs (Extended 2026 Edition Stage A+)*

8.2.2 Stage D Data Accumulation

The DELTA cycle initiates with data. An API-first payment platform, by virtue of processing every transaction on behalf of its merchant customers, accumulates a proprietary dataset of extraordinary richness and granularity. Razorpay processes approximately one billion transactions per quarter providing transaction-level detail across 100+ payment methods that is structurally superior to any external observer's view, including the merchant's own bank. The data accumulation stage is self-reinforcing: more merchants generate more transactions, which generate more data, which enable better products, which attract more merchants.

8.2.3 Stage E Embedded Finance Expansion

The transition from data accumulation to embedded finance is the model's most commercially consequential stage. Razorpay Capital's 10-second disbursal time (2026), Line of Credit up to Rs 25 lakh, and estimated NPA ratio of approximately 3.0% (vs 6.8% industry benchmark) achieved precisely because the underwriting model uses real-time transaction data unavailable to any lender who does not simultaneously process the merchant's payments is the empirical centerpiece of this stage.

8.2.4 Stage L Lock-in Creation

The lock-in stage describes the cumulative effect of deep product integration on merchant switching behavior. Razorpay's automation products generate three classic switching cost types simultaneously: procedural (workflow reconstruction cost), financial (sunk integration investment and data history value), and relational (mutual operational knowledge). Stage A+ (Agentic Intelligence) introduces a fourth: AI model calibration loss the operational intelligence that Razorpay's agents have accumulated about a specific merchant's dispute patterns, payment failure modes, and cash-flow seasonality cannot be transferred to a competing platform.

8.2.5 Stage T Technology Reinforcement

Revenue generated by embedded finance products and retention secured by lock-in creates the resource base for continued technology investment. For Razorpay, Stage T is visible in the Payment Optimizer ML model (15–20% payment success rate improvement), the final churn prediction model (ROC-AUC 0.967), the embedded finance adoption model (ROC-AUC 0.963), the 30+ e-commerce integrations, eight-language SDK coverage, biometric passkey authentication with MasterCard and Visa, and the AI infrastructure underpinning Agent Studio.

8.2.6 Stage A Advancement and Ecosystem Growth

The advancement stage describes the platform's progressive expansion into new merchant segments, geographies, and product categories. Razorpay's Stage A evidence includes: Malaysia (Curlec, now Razorpay); Middle East (2,000+ merchants, April 2026); UPI live in 25 countries (Israel added April 2026); Project Nexus multilateral linkage (2026); the confidential IPO SEBI filing (April 2026, USD 600–700M, USD 5–6B); and the Sarvam partnership enabling voice-first conversational commerce in Indian languages.

8.2.7 Stage A+ Agentic Intelligence (New 2026)

Stage A+ is proposed as a new sixth stage that has emerged in Razorpay's trajectory during 2025–2026. It represents the deployment of autonomous AI agents that transform accumulated transaction data and embedded financial relationships into an operational intelligence capability that continuously optimizes financial performance and deepens switching costs. Stage A+ is distinguished from Stage T (Technology Reinforcement) by its operational character: where Stage T improves product quality and reduces unit costs, Stage A+ replaces human operational processes with autonomous agents operating faster, more consistently, and at near-zero marginal cost.

Stage A+ is also distinguished from prior competitive mechanisms by the mechanism of switching-cost creation. Traditional API integration creates switching costs at the code level (estimated cost: weeks of engineering time). Agent Studio creates switching costs at the AI calibration level (estimated cost: months of agent training data and operational pattern learning). The MCP Server creates switching costs at the AI abstraction level (estimated cost: AI model re-prompting or re-training across an entire developer ecosystem). Each successive layer operates above the previous, suggesting that competitive moats in the agentic AI era may prove more durable than those in the API era.

8.3 Application of the DELTA Model to Razorpay (May 2026)

DELTA Stage	Razorpay Evidence (May 2026)	Stage Level	Key Updated Metrics
D Data	1 billion quarterly transactions; 12M+ merchants; 100+ payment methods; 7.4 billion+ total lifetime transactions; zero major data breaches	Advanced	USD 45B quarterly TPV; transaction-level data across 7+ years
E Embedded Finance	Razorpay Capital (10-sec disbursals, Rs 25L LoC, ~Rs 6,000 Cr annual est.); RazorpayX (40,000+ payroll	Advanced	NPA ~3.0% vs 6.8% industry; 94% merchant retention

DELTA Stage	Razorpay Evidence (May 2026)	Stage Level	Key Updated Metrics
	businesses); Opfin; InsurTech embedding		
L Lock-in	Smart Collect; Route; Payouts; DirectToPhone; Magic Checkout (22% cart abandonment reduction); Payments as a Service; Agent Studio AI calibration lock-in	Advanced	3–5% churn multi-product vs 15–22% single-product; 4.2 avg products enterprise
T Technology	Optimizer (15–20% success improvement); Churn model (ROC-AUC 0.967); embedded finance adoption model (ROC-AUC 0.963); 8-language SDK; biometric passkey auth; Sarvam voice commerce	Advanced	55% market share; Rs 3,783 Cr FY25 revenue (65% growth)
A Advancement	Malaysia (Curlec); Middle East 2,000+ merchants; UPI 25 countries (Israel Apr 2026); Project Nexus 2026; confidential IPO filing (Apr 2026, USD 5–6B)	Moderate-Advanced	USD 600–700M IPO target; 4,035 employees; USD 822M total funding
A+ Agentic Intelligence (New)	Agent Studio: Dispute Responder, Failed Payment Recovery, Cash-Flow Forecasting, No-Code Builder (March 2026); OpenAI Codex MCP Server (Apr 2026); Replit partnership; ChatGPT payments app	Early-Advancing	First mover; 1M+ developer weekly Codex sessions exposed to Razorpay MCP

Table 8.1 (Extended) *Application of DELTA Model to Razorpay Evidence Mapping (May 2026) (Sources: RoC FY25 Filings; NPCI 2026; Razorpay Press Releases)*

8.4 Originality, Theoretical Contribution, and 2026 Extension

The DELTA Model's original four contributions synthesis of four theoretical streams; explicit developmental sequencing; positive feedback mechanism modelling; and generalisability across API-first platforms are extended in 2026 by two additional contributions. Fifth: Stage A+ (Agentic Intelligence) is the first academic framework component to incorporate autonomous AI agents as a source of fintech competitive advantage, establishing a new theoretical strand at the intersection of agentic AI literature (Davidovic & Tourpe, 2026) and fintech competitive advantage theory. Sixth: MCP Server integrations are identified as a new mechanism of developer-side switching-cost creation operating at the AI abstraction level a mechanism not anticipated in any prior platform economics literature.

8.5 Limitations of the DELTA Model

Three limitations warrant acknowledgement. First, the model is inductively derived from a single case study, though partially corroborated by comparison with Stripe, Nubank, and Alipay. Multi-case empirical validation remains a primary future research priority. Second, Stage A+ is based on Razorpay's March–April 2026 deployments, which are too recent for outcome evaluation. Third, the MCP switching-cost mechanism is entirely novel and has no prior academic treatment; the claim that it creates more durable switching costs than SDK integration is theoretically plausible but empirically unvalidated.

CHAPTER 9

Strategic Recommendations

9.1 Recommendations for FinTech Firms

9.1.1 For Razorpay   Accelerating the DELTA Cycle (2026–2027 Priorities)

Razorpay's strategic priority for FY2026–27 should focus on five initiatives that build on its DELTA advantages:

- Maximize Account Aggregator Integration for Razorpay Capital: The AA Framework enables supplementing transaction-data underwriting with consented multi-source financial data   GST filings, bank statements across all merchant banking relationships, insurance premium histories. Priority investment should be directed at completing AA-compliant data ingestion pipelines to offer lower-cost, higher-ticket loans to merchants whose creditworthiness is strong but whose documentation has historically been inadequate for traditional NBFC approval.
- Scale Agent Studio with Governance Leadership: As the first large-scale agentic AI deployment in Indian payments, Razorpay has an opportunity to shape emerging regulatory frameworks. Proactive engagement with RBI and SEBI on agent governance standards   transparency requirements, escalation protocols, liability frameworks   converts regulatory risk into competitive positioning: governance-compliant agent deployment becomes a barrier that late-moving competitors face.
- Leverage IPO Capital for International DELTA Replication: The confidential IPO filing (April 2026, USD 600–700M) positions Razorpay to fund Stage A advancement at scale. Priority should be directed at Southeast Asia (building on Malaysia/Curlec) and the Middle East (from 2,000 merchants toward full-stack financial services), applying the sequenced DELTA Model logic: data accumulation first, embedded finance expansion second, lock-in creation third.

- Build an Embedded InsurTech Layer: Transaction data enables real-time insurance underwriting   a merchant processing Rs 50 lakh monthly has a verifiable revenue base supporting parametric business interruption insurance at actuarially accurate pricing. Initial targets: cyber fraud insurance, inventory protection, and professional liability distributed at point of transaction or account setup.
- Develop a Profitability Narrative for Post-IPO Credibility: FY25's Rs 3,783 crore revenue and Rs 1,277 crore gross profit demonstrate a credible path to operating profitability, though one-off costs (ESOP, domicile-flip) distorted net results. Priority: demonstrate unit economics improvement   specifically, margin improvement per active merchant as product depth increases   to build the investor narrative for the public listing.

9.1.2 For Emerging FinTech Start-ups   Navigating Razorpay's Market Dominance

Direct competition with Razorpay on its core competencies is inadvisable. Three alternative positioning strategies offer better risk-adjusted prospects:

- Vertical Embedded Finance Specialisation: Healthcare revenue cycle management, agricultural supply chain finance, and education fee collection represent segments where generic solutions leave significant friction. Sector-specific solutions can achieve deeper integration and data advantage within their vertical than a horizontal platform like Razorpay can justify.
- Cross-Border Payments Infrastructure: India's USD 776 billion goods and services export base is chronically under-served by efficient digital payment infrastructure. A specialist in B2B cross-border payments   particularly for MSME exporters   can build a defensible position in a segment where Razorpay has limited investment focus.
- MSME Financial Operating System: Building an integrated financial OS for MSMEs that uses Razorpay as a payment processing provider while owning the accounting, tax compliance, banking, and investment product relationships avoids direct gateway competition while capturing the most valuable parts of the merchant relationship.

9.2 Recommendations for Incumbent Banks

- Become Banking as a Service Infrastructure Providers: Develop BaaS offerings packaged APIs for account opening, fund transfer, KYC verification, balance inquiry, and regulatory reporting for fintechs to build on bank infrastructure rather than around it. HDFC SmartHub and Axis API products are early-stage models; scaling and standardizing this approach converts the fintech displacement threat into cooperative infrastructure revenue.
- Develop AI Agent Governance Expertise: As Razorpay's Agent Studio scales, banks that have invested in understanding agentic AI governance   transparency requirements, liability frameworks, escalation protocols   will be better positioned to partner with, regulate, and ultimately co-deploy AI-powered financial operations.
- Engage Proactively in the Account Aggregator Ecosystem: Banks that become active, high-quality data contributors to the AA Framework position themselves as preferred financial data sources for fintech underwriting engines including Razorpay Capital   converting the fintech threat into a data partnership revenue opportunity.
- Acquire or Partner for Technology Capabilities: The most efficient path to competitive AI/ML underwriting capabilities and developer ecosystem relationships is acquisition or deep partnership with fintechs that already possess them, prioritizing companies that extend the bank's capabilities without cannibalizing core deposits and lending business.

9.3 Recommendations for SME Business Users

- Strategically Adopt Multi-Product Depth: The data demonstrates that merchants using three or more Razorpay products achieve substantially better cash-flow management, lower operational costs, and higher payment success rates. Approach the Razorpay relationship as a strategic financial infrastructure decision rather than a transactional tool selection.
- Leverage Razorpay Capital and the Line of Credit Proactively: The 10-second disbursal capability and Line of Credit (up to Rs 25 lakh at 1.5%/month, no

collateral) are most valuable when used proactively   to fund working capital ahead of festive season inventory builds, bridge receivables gap during growth phases, or finance equipment investments.

- Adopt Agent Studio for Operational Efficiency: For merchants with meaningful dispute volume or complex payment settlement patterns, the Agent Studio Dispute Responder and Failed Payment Recovery agents offer tangible operational cost reduction and cash-flow improvement. Early adoption builds AI calibration depth that improves outcomes over time.
- Participate in the Account Aggregator Framework: SMEs that consent to share multi-institution financial data will access more accurate credit assessments, potentially more favorable loan pricing, and faster approvals   with the commercial incentive to participate clear for businesses whose transaction data alone understates their creditworthiness.

9.4 Recommendations for Regulators

- Develop an AI Agent Governance Framework for Payments: The deployment of agentic AI in payment operations   where agents autonomously initiate, execute, and reconcile financial transactions   is not addressed by any existing Indian regulatory framework as of May 2026. The RBI and SEBI should develop a dedicated governance framework addressing: accountability (liability for agent errors), transparency (merchant disclosure requirements), and systemic risk (correlated agent behavior during economic stress).
- Accelerate Account Aggregator Ecosystem Maturation: Mandate high-quality data provision from banks above a specified asset threshold; develop common data format standards; invest in consumer awareness programmers. The AA Framework's potential to enable data-driven financial inclusion at scale remains under-realized due to implementation friction.
- Develop a Graduated FinTech Lending Licence Regime: The current NBFC framework imposes the same compliance burden on small-ticket gig-economy lenders as on large corporate credit intermediaries. A graduated licensing framework   calibrated to loan ticket size, borrower type, and portfolio scale

would lower barriers for responsible small-ticket embedded lenders while maintaining appropriate oversight at scale.

- Mandate Bank API Interoperability Standards: A mandate for all banks above a specified size to publish standardised APIs for core banking functions would create infrastructure for the next wave of embedded financial services replicating the success of UPI's mandatory interoperability in the payment layer at the banking API layer.
- Establish a Systemic Risk Monitoring Framework for Fintech and AI Lending: As embedded fintech lending and agentic AI operations scale, the RBI should develop real-time reporting for large fintech platforms covering portfolio concentration, algorithmic underwriting model similarity, credit limit adjustment patterns, and AI agent operational metrics enabling early identification of correlated behaviour that could amplify macroeconomic stress.

9.5 Implementation Roadmap (2026–2027)

Recommendation	Stakeholder	Priority	Horizon	Feasibility
Scale Agent Studio with regulatory engagement	Razorpay	High	6–12 months	High first-mover advantage window
IPO capital deployment for SEA/ME expansion	Razorpay	High	12–24 months	Medium IPO timeline uncertain
Accelerate AA integration for Capital underwriting	Razorpay	High	6–12 months	High technical work advanced
Build embedded InsurTech layer	Razorpay	Medium	18–30 months	Medium requires insurance partnerships
BaaS API development and fintech partnerships	Incumbent Banks	High	18–24 months	Medium significant cultural investment
AI agent governance framework development	RBI/SEBI	High	12–18 months	Medium policy development complex

Recommendation	Stakeholder	Priority	Horizon	Feasibility
Graduated FinTech lending licence regime	Regulators	High	18–24 months	Medium FCA model provides template
Bank API interoperability mandate	Regulators	High	24–36 months	Low-Medium significant bank resistance
Proactive multi-product Razorpay adoption	SMEs	High	0–6 months	High low friction
Agent Studio adoption for operational efficiency	Enterprise SMEs	Medium	6–12 months	High commercially incentivized
AA consent framework participation	SMEs	Medium	6–12 months	High voluntary; commercially incentivized

Table 9.2 Implementation *Roadmap: Priority, Timeline, and Feasibility Assessment (2026–2027)*

CHAPTER 10

Conclusion

10.1 Summary of Research Contributions

This dissertation set out to investigate financial services innovation in India through both the wide-angle lens of macro-structural forces and the focused lens of Razorpay as an empirical case study, updated to incorporate the most consequential developments through May 2026.

It has synthesised seven bodies of academic and practitioner literature financial innovation theory, disruptive innovation, platform economics, digital payments, embedded finance, fintech regulation, and added in this 2026 edition agentic AI in financial services into a coherent analytical framework for understanding how API-first fintech companies build competitive advantage in developing market contexts.

It has analysed Razorpay's twelve-year evolution from a developer-friendly payment gateway (2014) to an AI-powered full-stack financial services platform (2026), serving 12 million+ businesses, processing USD 180 billion in annual TPV, generating Rs 3,783 crore in FY25 revenue with 65% YoY growth, and positioning for a USD 5–6 billion public market debut.

It has introduced and updated the DELTA Model, extending it with Stage A+ (Agentic Intelligence) to reflect the qualitative shift from passive infrastructure to active financial intelligence, and has identified MCP Server integrations as a new platform economics mechanism of developer-side switching-cost creation operating at the AI abstraction level.

10.2 Answer to Research Questions

Research Question	Summary Answer (Updated May 2026)
RQ1: Structural forces driving innovation 2014–2026?	UPI's 22.35B monthly transactions (April 2026) representing 50% of global real-time digital transaction volume; Aadhaar-enabled digital KYC; MSME GST formalisation; COVID-19 digitalisation; DPDP Act 2025; Digital Lending Directions 2025; RBI cross-border PA approval; and from 2025–2026 agentic AI as an infrastructural innovation force without precedent.
RQ2: How has Razorpay's API-first strategy generated competitive advantages?	Through five mechanisms: (1) API quality creating switching costs and developer network effects; (2) transaction data enabling superior embedded finance underwriting; (3) automation products embedding operational integration; (4) regulatory compliance creating institutional legitimacy; (5) from 2026: Agent Studio agentic AI creating calibration-level switching costs above the code layer.
RQ3: What are the impacts of Razorpay Capital on SME credit access?	Razorpay Capital's 10-second disbursal (vs 4–8 weeks NBFC), Line of Credit up to Rs 25 lakh, estimated NPA ~3.0% (vs ~6.8% industry benchmark) achieved because real-time transaction underwriting is structurally superior to periodic financial statement assessment demonstrates that India's USD 530 billion SME credit gap is a measurement failure, not a genuine risk failure.
RQ4: Can the extended DELTA Model explain Razorpay's trajectory?	Yes. All six DELTA stages are documented with evidence: D (advanced), E (advanced), L (advanced), T (advanced), A (moderate-advanced), and A+ (early-advancing). The DELTA cycle is demonstrably reinforcing and accelerating, with Stage A+ representing the highest-impact development in the company's history from a competitive advantage perspective.
RQ5: What strategic responses are available to stakeholders?	For banks: BaaS repositioning and AA ecosystem participation. For fintechs: vertical specialisation, cross-border focus, or MCP-enabled AI-native product building. For SMEs: proactive multi-product adoption and Agent Studio operational efficiency. For regulators: AI agent governance framework development, graduated lending licensing, and bank API interoperability mandate with urgency increased by the pace of agentic AI deployment.

Table *Summary: Research Questions and Answers (Updated May 2026)*

10.3 Broader Implications for Financial Services

Beyond Razorpay specifically, this dissertation's findings carry three broader implications for the future of financial services.

First, the DELTA Model's Stage A+ suggests that agentic AI is not a feature to be added to fintech products but a new competitive dimension that will define the next era of financial services competition. Companies that have accumulated sufficient transaction data (Stage D) and deployed AI infrastructure (Stage T) are positioned to enter Stage A+ rapidly; those that have not will find that the switching costs of the agentic AI era are the most durable in fintech history.

Second, the embedded finance findings particularly Razorpay Capital's 10-second disbursals and NPA performance consistently below industry benchmarks provide evidence that structured transaction data can reduce information asymmetry in SME credit. Embedded finance is therefore treated as a mechanism for liquidity visibility and risk-based credit targeting.

Third, India's regulatory approach enabling innovation through infrastructure investment, calibrating risk through proportionate licensing, and progressively expanding data sharing through the Account Aggregator Framework represents a model that merits study by regulatory authorities in comparable developing economies. The emerging question for the agentic AI era is whether India can extend this regulatory leadership to govern autonomous AI financial agents before the risks they introduce correlated algorithmic behaviour, accountability gaps, systemic liquidity effects become material.

10.4 Limitations and Future Research Directions

This dissertation has several limitations that future research should address. The DELTA Model requires validation across multiple additional cases Stripe, Nubank, Adyen, GrabPay, Flutterwave before its generalisability claims can be established empirically. Stage A+ is based on fewer than three months of production data from Agent Studio and the OpenAI Codex integration; rigorous evaluation of agentic AI as a competitive advantage mechanism requires 12–24 months of operational data.

Future research directions include: a primary research study of enterprise SME experiences with agentic AI adoption (Agent Studio outcomes, MCP integration utility); a longitudinal study of Razorpay Capital's NPA performance across economic cycles; a comparative regulatory analysis of India's fintech policy architecture against Brazil's Open Finance and the UK's Open Banking standard; an empirical test of MCP-mediated switching costs in AI-mediated software development; and a multi-case DELTA Model application to five global API-first fintech platforms.

10.5 Final Reflection

The story of Razorpay is ultimately a story about information specifically, about what becomes possible when the information required to assess creditworthiness, manage cash flows, process payrolls, execute vendor payments, manage disputes, and forecast liquidity is made programmable accessible through APIs, and now autonomously actionable through AI agents, rather than locked within siloed institutional databases managed by human operators.

By April 2026, Razorpay had embedded UPI into an AI coding platform used by over a million developers weekly, enabling any developer to create revenue-generating payment-enabled applications in under five minutes. Its autonomous AI agents manage financial operations for millions of merchants without human initiation at the individual task level. These developments do not merely represent the latest chapter in India's digital finance revolution they represent a preview of how financial services will be delivered globally in the decade ahead.

The DELTA Model, the empirical Razorpay case, and the five research questions addressed in this dissertation are offered not as the final word on this story but as a rigorous analytical foundation updated to May 2026 for the conversations in which the story's next chapter will be shaped: by regulators, entrepreneurs, bankers, merchants, and the AI systems that, increasingly, act on their behalf.

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All references are presented in APA 7th edition format, listed in alphabetical order, and updated to May 2026. Single-spaced with hanging indent. Where earlier editions have been superseded, the current version is cited.

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CHAPTER

Appendix 1: Power BI Dashboard Design and Analytical Framework

This appendix documents the architecture, data model, and key visualisations of the Power BI dashboard suite developed in support of this dissertation's quantitative analysis. All dashboards are designed for operational use by a fintech company's CFO and data analytics team.

A.1 Dashboard Suite Overview

Dashboard	Primary Audience	Refresh Frequency	Key Purpose
Executive Summary	CEO, CFO, Board	Daily	TPV, merchant count, success rate, NPA top-line health at a glance (updated Apr 2026 data)
Merchant Analytics	VP Sales, Account Management	Daily	Merchant segment performance, churn risk scores, cross-sell propensity, Agent Studio adoption
Lending Portfolio	CFO, Risk Manager	Daily	Razorpay Capital portfolio quality, NPA trends, 10-sec disbursal KPIs
Operational Efficiency	COO, Finance Operations	Weekly	Settlement lag, Smart Collect automation rate, payout SLA, Agent Studio recovery rate
Competitive Benchmarking	Strategy, Business Development	Monthly	Payment success rates, MDR comparison, product feature benchmarking vs Stripe/Cashfree
AI Agent Performance (New 2026)	CFO, Head of AI	Daily	Dispute win rate (Dispute Responder); failed payment recovery rate; cash-flow forecast accuracy (Agent Studio)

Table A.1 *Power BI Dashboard Suite Overview (Updated May 2026)*

A.2 Data Architecture Star Schema Model

The data model follows a standard star schema design with one central fact table and six dimension tables:

- FactTransactions: Central fact table. Key fields: transaction_id, merchant_id, payment_method_id, date_key, amount_inr, success_flag, MDR_charged_inr, settlement_date, settlement_lag_days, agent_studio_flag.
- DimMerchant: Merchant master data. Key fields: merchant_id, merchant_name, industry_segment, GMV_tier, onboarding_date, products_active, KYC_status, account_manager_id, agent_studio_adopted.
- DimDate: Full calendar table. Key fields: date_key, date, month, quarter, fiscal_year, is_festive_season, is_fiscal_year_end.
- DimPaymentMethod: Payment method reference. Key fields: method_id, method_name, method_category (UPI/Card/NetBanking/Wallet/BNPL/Other), is_zero_MDR.
- DimLoan: Lending portfolio dimension. Key fields: loan_id, merchant_id, loan_amount, disbursement_date, outstanding_principal, NPA_flag, disbursal_seconds, loan_tenure_months, underwriting_model_version.
- DimProduct: Product adoption dimension. Key fields: merchant_id, product_name, adoption_date, product_category, subscription_tier.
- DimAgentStudio (New 2026): Agent performance dimension. Key fields: agent_id, agent_type (Dispute/Recovery/Forecast), merchant_id, case_id, action_taken, outcome_flag, resolution_seconds.

A.3 Core DAX Measures

Measure Name	DAX Logic Description	Business Purpose
Total TPV	SUM of transaction amounts for all captured transactions	Primary business scale metric
Payment Success Rate	DIVIDE(SUM captured, COUNT all transactions)	Infrastructure quality indicator
Active Merchants MTD	DISTINCTCOUNT merchant_id with ≥1 transaction in current month	Growth and engagement metric

Measure Name	DAX Logic Description	Business Purpose
MDR Revenue	SUM MDR_charged_inr across all transactions	Primary gateway monetisation measure
GMV MoM Growth %	DIVIDE(Current month GMV minus Prior month GMV, Prior month GMV)	Momentum indicator
NPA Ratio	DIVIDE(SUM outstanding where NPA_flag=TRUE, SUM total outstanding)	Portfolio credit quality metric
Disbursal Time Avg (Seconds)	AVERAGE of disbursal_seconds from DimLoan	Capital product speed KPI target <15 seconds
Agent Dispute Win Rate	DIVIDE(COUNT outcomes where outcome_flag=WIN, COUNT all disputes)	Agent Studio Dispute Responder performance
Failed Payment Recovery Rate	DIVIDE(TPV recovered by agent, TPV submitted to recovery queue)	Agent Studio Recovery performance
Churn Risk Score (Avg)	AVERAGE of merchant_churn_score from AI model output table	Retention risk monitoring
UPI Share	DIVIDE(TPV where method_category=UPI, Total TPV)	Payment method mix monitoring
Cross-Sell Propensity (High)	COUNT merchants where adoption_propensity_score > 0.65	Product specialist outreach prioritisation

Table A.2 *Core Power BI DAX Measures: Definitions and Business Purpose (Updated May 2026)*

A.4 Dashboard Page Descriptions

Page 1 Executive Summary

KPI card row (top): Total TPV (month-to-date), Payment Success Rate, Active Merchants MTD, MDR Revenue, Average Settlement Lag Days, Agent Studio Cases Resolved. Each card shows current period value, prior period comparison, and trend arrow. Main visual panel: 12-month TPV trend line with month-on-month growth percentage annotation. Secondary panel: Payment composition donut (UPI, Cards, Net Banking, Wallets, BNPL, Other). Geographic panel: Filled map of India showing state-level TPV concentration.

Page 2 Merchant Analytics

Merchant segment bar chart showing TPV, merchant count, and average products used. Scatter plot: Merchant TPV vs Payment Success Rate identifies high-value merchants with suboptimal success rates who are Optimizer upgrade candidates. Churn risk heatmap by segment and industry vertical. Agent Studio adoption tracker (% of enterprise and growth-stage merchants with at least one active agent).

Page 3 Lending Portfolio

KPI cards: Total Disbursements (MTD), Portfolio Outstanding, NPA Ratio, Average Loan Size, Average Disbursal Time (seconds). Waterfall chart: portfolio movement (opening, new disbursements, repayments, prepayments, write-offs, closing). NPA trend line with 3.5% governance target. Loan ticket-size distribution. Disbursal time trend (showing reduction from hours to seconds over 2023–2026).

Page 4 Operational Efficiency

Smart Collect automation rate trend (target 85%+). Settlement lag distribution. Payout SLA compliance gauge (target 98%+ within 4 minutes). Payment failure rate heatmap by method and acquiring bank.

Page 5 Agent Studio Performance (New 2026)

Dispute Responder: Monthly case volume, win rate trend, average resolution time (seconds vs prior manual baseline). Failed Payment Recovery: Weekly recovery queue volume, recovery rate %, TPV recovered (Rs crore). Cash-Flow Forecasting: Forecast accuracy metric (MAPE Mean Absolute Percentage Error vs actual settlement outcomes). No-Code Builder: Number of custom agents created, usage frequency, outcome distribution.

CHAPTER

Appendix 2: AI/ML Model Design Summary and Business Application

B.1 Merchant Churn Prediction Model Summary

Dimension	Description
Model Type	Gradient-boosted decision tree (XGBoost) with SHAP interpretability layer
Prediction Target	Binary: Will this merchant reduce TPV by >70% within next 90 days?
Training Data	24 months of historical transaction and product usage data from representative cohort
Top Predictive Variables	(1) 30-day TPV velocity change; (2) Products active; (3) 30-day payment success rate; (4) Open support tickets >7 days; (5) Settlement lag complaints; (6) Agent Studio adoption status (new 2026)
Model Performance	FINAL VALIDATION ROC-AUC 0.967; Precision 1.000, Recall 0.927, F1 0.962
Deployment Context	Batch scoring run nightly; outputs integrated into CRM as 'Churn Risk Score' field
Business Action	Score >0.7: Immediate AM outreach with personalised retention playbook; Score 0.5–0.7: Automated in-app recommendation; Score <0.5: Standard engagement
Estimated Annual Value	INR 32+ crore incremental revenue preservation (40% retention rate of at-risk merchants)

Table B.1 Merchant *Churn Prediction Model: Summary Specification (Updated with Agent Studio variable, 2026)*

B.2 Embedded Finance Adoption Propensity Model Summary

Dimension	Description
Model Type	Logistic regression with L2 regularisation (Ridge) for interpretability
Prediction Target	Probability merchant will adopt RazorpayX banking products within next 6 months
Training Data	18 months of product adoption history + transaction behaviour features
Top Predictive Variables	(1) Monthly vendor payout volume through non-Razorpay channels; (2) Daily settlement frequency; (3) GST tool active usage (2.8x higher propensity); (4) Merchant tenure 12–36 months; (5) Industry: marketplace/SaaS vs retail
Model Performance	FINAL VALIDATION ROC-AUC 0.963; Precision 0.892, Recall 0.840, F1 0.866
Deployment Context	Monthly scoring batch; used by product specialist team for cross-sell prioritisation
Business Action	Score >0.65: Product specialist direct outreach with payout audit offer; Score 0.4–0.65: In-dashboard trial prompt; Score <0.4: Educational content
Estimated Annual Value	INR 18–25 crore incremental RazorpayX revenue from model-prioritised cross-sell

Table B.2 *Embedded Finance Adoption Propensity Model: Summary Specification*

B.3 Model Governance Framework (Updated for DPDP Act 2025)

- Fairness Monitoring: Both models are tested quarterly across merchant segments, industry verticals, and geographic regions. DPDP Act 2025 Section 4 purpose limitation compliance verified. No demographic characteristics (caste, religion, gender, geography) are used as primary inputs.
- Drift Detection: AUC, precision, and recall monitored monthly on rolling validation set. Performance degradation >5 percentage points triggers re-training review.
- Explainability Standard: Every model output used in a merchant-facing decision must have a human-interpretable SHAP explanation available on request,

consistent with DPDP Act 2025 automated decision-making disclosure requirements.

- Human Override: Account managers retain authority to override churn risk classifications. Override rates and outcomes are tracked to assess model calibration quality.
- Agent Studio Governance (New 2026): Anthropic Claude SDK guardrails enforce action boundaries. All agent decisions above defined monetary thresholds (Rs 10,000 dispute value; Rs 50,000 recovery amount) require human escalation. Agent action logs are maintained for audit per RBI PA licence renewal conditions.

CHAPTER

Appendix 3: Dataset and Industry Data Reference

C.1 Primary Dataset Descriptions

Dataset	Source	Coverage	Primary Use in Study
India UPI Transaction Statistics	NPCI Monthly Data Publications	FY2019–April 2026 (monthly)	Chapters 1, 4, 5: Payment trends and market context
India FinTech Market Sizing	IMARC Group 2026	2021–2034 (annual + forecast)	Chapters 1, 4: Market size and growth trajectory
Razorpay Performance Metrics	RoC FY25 Filings; Inc42; Entrackr; Bloomberg; Press Releases	FY2020–FY2026 (estimated/disclosed)	Chapters 4 and 5: Company analysis
Global FinTech Investment Data	Innovate Finance 2025; KPMG Pulse of FinTech	2015–2025 (annual)	Chapter 1: Global context
India NBFC SME Lending Data	RBI Report on Trend and Progress of Banking	FY2021–FY2024 (annual)	Chapter 5: Razorpay Capital benchmarking
Agent Studio Performance	Razorpay Blog (March 2026); Press Releases	March–May 2026	Chapter 6: Agentic AI analysis (exploratory)
UPI Monthly Stats (Apr 2026)	NPCI, May 4 2026	April 2026 (monthly)	Chapter 1, 5: Latest UPI data anchor
Competitor Data	Stripe/Adyen annual reports; PayU Prosus filings	2022–2025	Chapter 4: Competitive benchmarking
Embedded Finance Market	Research and Markets Q4 2025	2021–2030	Chapter 2: Literature review market sizing

Dataset	Source	Coverage	Primary Use in Study
India Digital Lending	RBI Digital Lending Directions 2025	FY2022–2026	Chapters 4 and 9: Regulatory analysis

Table C.1 Primary *Datasets: Sources and Application (Updated May 2026)*

C.2 Global FinTech Ecosystem Comparative Reference

Dimension	India (Razorpay)	USA (Stripe)	UK (Starling/Monzo)	China (Alipay)	Brazil (Nubank)
Payment Rail	UPI (govt-built, zero MDR; 22.35B monthly Apr 2026)	Card networks (Visa/MC)	Faster Payments	Proprietary super-app	PIX (Banco Central)
Dominant Mode	UPI / QR Code (82% volume share)	Credit/Debit Card	Bank Transfer	QR Wallet	PIX Instant Transfer
Embedded Finance	Advanced (Capital 10-sec disbursals; RazorpayX; InsurTech)	Advanced (Stripe Capital)	Marketplace model	Advanced (Ant Group)	Advanced (NuLend)
Agentic AI Layer	Yes Agent Studio (March 2026, Anthropic Claude SDK)	Limited public disclosure	No	Partial (Ant AI)	No
MCP / AI Integration	Yes OpenAI Codex (April 2026), first payments MCP globally	In development	No	No	No
Regulatory Model	RBI Calibrated innovation; DL Directions 2025; DPDP 2025	OCC/Fed/CFPB	FCA Sandbox-led	PBOC Tight control	BCB Open Finance
SME Credit Gap	Large (USD 530B); being addressed by	Moderate	Moderate	Partially addressed	Large

Dimension	India (Razorpay)	USA (Stripe)	UK (Starling/Monzo)	China (Alipay)	Brazil (Nubank)
	Capital (10-sec, Rs 25L LoC)				
DELTA Stage (May 2026)	D+E+L+T+A+A+ (first mover A+)	D+E+L+T+A (all 5)	D+E+L	D+E+L+T+A (all 5)	D+E+L+T (A emerging)
IPO/Listing Status	Confidential SEBI filing April 2026 (USD 5–6B)	Private (USD 65B valuation)	Starling: IPO shelved 2023	Ant: IPO blocked by PBOC	Listed NYSE (2021)

Table C.2 Global *FinTech Ecosystem Comparison with DELTA Stage Assessment (Updated May 2026)*

C.3 Razorpay API Catalogue Key Endpoints (Updated)

API Category	Key Endpoints	Integration Scope	2026 Addition
Payment Acceptance	/orders, /payments, /checkout, /payment-links, /qr-codes	100+ payment methods; 8-language SDK; 30+ e-commerce integrations	Q-Zap offline POS; Payments as a Service
Payouts and Banking	RazorpayX: /payouts, /virtual-accounts, /contacts, /fund-accounts	DirectToPhone; Forex/FDI; Escrow+; tax payments	DirectToPhone Payouts (phone number only)
Lending	Razorpay Capital: /loans, /line-of-credit, /underwriting	NBFC-partner model; AA Framework integration	10-second disbursal; Rs 25L Line of Credit
Marketplace	Route: /transfers, /settlements, /refunds	Real-time split payments; automated vendor settlement	Enhanced multi-party settlement rules
AI / Agentic	Agent Studio: /agents/dispute, /agents/recovery, /agents/forecast	Anthropic Claude SDK; guardrail-enforced actions	MCP Server for OpenAI Codex (April 2026)
Authentication	/tokens, /mandates, /recurring	UPI Autopay; eMandate; biometric passkey	Mastercard/Visa biometric passkey (2025–2026)

Table C.3 *Razorpay API Catalogue: Key Endpoints (Updated May 2026)*

Supervisor Evidence Alignment Addendum

Sections modified: Abstract, Research Objectives, Significance of Study, Chapter 3 Methodology, Chapter 6 AI/ML model metrics, Chapter 8 DELTA technology evidence, Chapter 10 Conclusion, Appendix AI/ML model metrics.

Reason for modification: Final outputs show strong churn and adoption models, a strong growth regression requiring churn-adjusted interpretation, and a default model that is useful for risk screening but precision-constrained.

Alignment summary: Proposal and report are aligned on Razorpay's platform evolution, DELTA framework, UPI context, and embedded finance strategy. The conclusion is revised so embedded finance is presented as supporting resilience, retention, liquidity stability, and profit continuity before measurable income expansion.

Key model evidence: Churn ROC-AUC 0.967, precision 1.000, recall 0.927, F1 0.962. Adoption ROC-AUC 0.963, precision 0.892, recall 0.840, F1 0.866. Default ROC-AUC 0.706 with low precision. Use for risk bands and human review. Income growth model R2 0.823 with churn-adjusted interpretation required.

Final conclusion: Growth appears constrained by churn and provider inactivity rather than technology failure. Embedded finance and digital adoption should be treated as operational enablers of retention, liquidity, and resilience.

Submission readiness score after report alignment: 86/100.